# 第1章

- (真空中の) 電磁場　　$\omega \propto k$ をみたす：電磁波
- 電磁場の力学
- 運動方程式 = Maxwell's eq.

$$\begin{cases} \dfrac{d\mathbb{X}}{dt} = \dfrac{1}{m}\mathbb{P} & \mathbb{X}(t) \\ \dfrac{d\mathbb{P}}{dt} = f(\mathbb{X}) & \mathbb{P}(t) \end{cases} \qquad \begin{cases} \dfrac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{j} & \text{連続方程式} \\ \dfrac{\partial \mathbf{j}}{\partial t} = -\nabla P(\rho) & \text{オイラー方程式} \end{cases}$$

圧力

$$\begin{cases} \mathbb{E}(\mathbb{X}, t) \\ \mathbb{B}(\mathbb{X}, t) \end{cases} \quad \text{eq. of motion ?} \qquad \begin{cases} \dfrac{\partial \mathbb{E}}{\partial t} = \boxed{?} \\ \dfrac{\partial \mathbb{B}}{\partial t} = \boxed{?} \end{cases}$$

axial vector

← $\mathbb{E}$, $\mathbb{B}$ およびその空間微分 ) 一次のベクトル

↓

真空中の重ね合わせ (Superposition) ⟶ 線型方程式　　rot しかありえない

polar $\nabla \times \mathbb{E}$

等方的 (isotropic)

局所的 (local) ⟶ 有限階数

一様な場は時間変化しない ⟶ $\mathbb{E}$, $\mathbb{B}$ の項は無い

$$\begin{cases} \dfrac{\partial \mathbb{E}}{\partial t} = a\, \nabla \times \mathbb{B} \\ \dfrac{\partial \mathbb{B}}{\partial t} = -b\, \nabla \times \mathbb{E} \end{cases}$$

カ $\mathbb{J}$ が付加わる.

$$\frac{\partial \mathbb{E}}{\partial t} = c^2 \nabla \times \mathbb{B} + \beta \mathbb{J} \qquad ab = c^2$$

$$\frac{\partial}{\partial t}(\nabla \cdot \mathbb{E}) = \beta \nabla \cdot \mathbb{J} \longrightarrow \frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J} \quad (\alpha\beta = c^2)$$

$\nabla \cdot \mathbb{E} = \alpha \rho(\mathbb{X}, t)$

magnetic monopole $\rho_m, \mathbf{J}_m = 0$ ⟶ $\nabla \cdot \dfrac{\partial \mathbb{B}}{\partial t} = \dfrac{\partial}{\partial t}(\nabla \cdot \mathbb{B}) = 0$

Lorentz force

点電荷　$\mathbb{F} = q[\mathbb{E}(\mathbb{X}, t) + \mathbb{v} \times \mathbb{B}(\mathbb{X}, t)]$

$\mathbb{P}$ ⟷ $\mathbb{E}$ polar

$\mathbb{X}$ ⟷ $\mathbb{B}$ axial

$$L = \frac{1}{8\pi}(\mathbb{E}^2 - \mathbb{B}^2)$$

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# 電磁場と電荷

e, p　　素電荷 $|e| = 1.6 \times 10^{-19}$ Coulomb ( = A·sec )

$q$ 点電荷に働く力 = Lorentz force　　$q \to$ 無限小 : test charge

$$\mathbf{F} = q\{\mathbf{E}(\mathbf{x},t) + \mathbf{v} \times \mathbf{B}(\mathbf{x},t)\}$$

$$P = mv/\sqrt{1 - v^2/c^2} \sim mv \quad (N.R)$$

## 第2章

時間変化しない ( 静的 static )

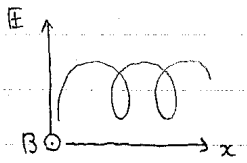
一様 ( uniform )

a)　$\mathbf{E} = 0$,　$\mathbf{B} = B\hat{z}$ (const)



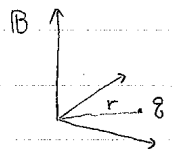
$\omega_c = \frac{q}{m}B$ : Cyclotron freq

b)　$\mathbf{E} \perp \mathbf{B}$



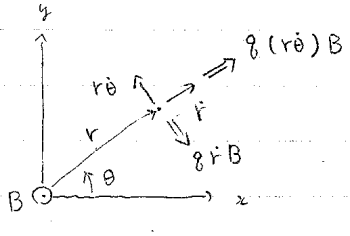

$\to$ Drift　　$v_D = E/B$

(問題) 調和振動子 ( 2次元 )




$\mathbf{E} = 0$

ポテンシャル　$V = \frac{m}{2}\omega_0^2 r^2$




$$\omega_c = \frac{qB}{m}$$

$$\begin{cases} m(\ddot{r} - r\dot{\theta}^2) = qBr\dot{\theta} - mr\omega_0^2 \\ m\frac{d}{dt}(r^2\dot{\theta}) = -qB\dot{r}\cdot r \end{cases}$$

$$\to \begin{cases} \ddot{r} - r\dot{\theta}^2 = \omega_c r\dot{\theta} - r\omega_0^2 \to \ddot{r} = -\left(\frac{1}{4}\omega_c^2 + \omega_0^2\right)r \\ r^2\dot{\theta} = -\frac{1}{2}\omega_c r^2 + \underset{\parallel}{\text{const}}_{0} \to \dot{\theta} = -\frac{1}{2}\omega_c \end{cases}$$

$$\Omega = \sqrt{\omega_0^2 + \frac{1}{4}\omega_c^2}$$

$$m\vec{v} = \left(\vec{p} - \frac{q}{c}\vec{A}\right)$$

$$\begin{cases} \nabla\cdot\mathbf{E} = \frac{1}{\epsilon_0}\rho & (\mathrm{I}) \\ \nabla\times\mathbf{E} + \frac{\partial\mathbf{B}}{\partial t} = 0 & (\mathrm{II}) \\ \nabla\times\mathbf{B} = \mu_0\left(\mathbf{j} + \epsilon_0\frac{\partial\mathbf{E}}{\partial t}\right) & (\mathrm{III}) \\ \nabla\cdot\mathbf{B} = 0 & (\mathrm{IV}) \end{cases} \xrightarrow{\text{static}} \begin{cases} \nabla\cdot\mathbf{E} = \frac{1}{\epsilon_0}\rho \\ \nabla\times\mathbf{E} = 0 \\ \nabla\times\mathbf{B} = \mu_0\mathbf{j} \\ \nabla\cdot\mathbf{B} = 0 \end{cases}$$

## §1 Coulomb の法則

Cavendish 1773

Coulomb 1785

action at distance $\quad \mathbf{F}_{12} = k q_1 q_2 \dfrac{\mathbf{x}_1 - \mathbf{x}_2}{|\mathbf{x}_1 - \mathbf{x}_2|^3}$

8. 正,負 neutral

$k = \dfrac{1}{4\pi\epsilon_0} = 10^{-7} c^2$

$\rho(\mathbf{x}) = \sum q_i \delta(\mathbf{x} - \mathbf{x}_i)$

$\mathbf{F} = q\mathbf{E}$

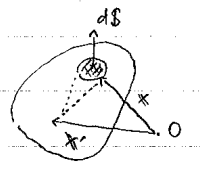
$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\int d\mathbf{V}'\rho(\mathbf{x}')\frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3} \quad (1) \xrightarrow{\text{微分形}} (\mathrm{I})$$

閉曲面 S でかこまれた体積 $V_S$ で (I) を積分

$$\int_S \mathbf{E}\, d\mathbf{S} = \frac{1}{\epsilon_0}\int_{V_S} d\mathbf{V}\,\rho(\mathbf{x})$$

(1) の面積分

$$\int d\mathbf{S}\cdot\frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3} = \begin{cases} 4\pi & \mathbf{x}' \text{ が S の内} \\ 0 & \text{〃 外} \end{cases}$$




(問題)

$$-\nabla\frac{1}{|\mathbf{x}-\mathbf{x}'|} = \frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3}$$

$$-\nabla^2\frac{1}{|\mathbf{x}-\mathbf{x}'|} = 4\pi\delta(\mathbf{x}-\mathbf{x}')$$

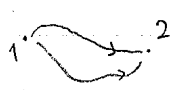
hint $\quad \lim_{\delta\to 0}(r^2+\delta^2)^{-\frac{1}{2}} = r^{-1}$

Earnshaw の定理 : 静電場だけでは平衡点はない

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## §2 電場のポテンシャル (potential)

(II) → 


$\int_1^2 \mathbf{E}\,d\mathbf{l}$ = 経路によらない = $q[\varphi(\mathbf{x}_2) - \varphi(\mathbf{x}_1)]$

$\therefore \mathbf{E}(\mathbf{x}) = -\nabla\varphi(\mathbf{x})$ $\varphi$: scalar potential

$$\varphi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\,dV'$$

静電場のエネルギー (自己エネルギーは除く)

$$\mathcal{E} = \frac{1}{2}\int \rho(\mathbf{x})\,\varphi(\mathbf{x})\,dV$$

$$\mathcal{E} = \frac{\epsilon_0}{2}\int \mathbf{E}^2\,dV$$

(問題)

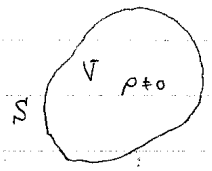
半径 $a$ の球, 内に一様に電荷分布 (total $q$)

$$\epsilon\int \mathbf{E}\,d\mathbf{S} = \int \rho\,dV \quad \rightarrow \quad \epsilon\,E(r)\cdot 4\pi r^2 = \begin{cases} \rho\cdot\frac{4}{3}\pi r^3 = q\left(\frac{r}{a}\right)^3 \\ q \end{cases}$$

$$\rightarrow \quad E(r) = \begin{cases} \frac{1}{4\pi\epsilon}\cdot\frac{qr}{a^3} \\ \frac{1}{4\pi\epsilon}\cdot\frac{q}{r^2} \end{cases}$$

# 導体 Conductor

- 内部で $E=0$
  Screening 遮へい効果
- 表面だけに $\rho$ がある。
  $\varphi = \text{const}$ $\quad E \parallel \hat{n}$




## 解の一意性 (Uniqueness)

S上でどんな条件？

2つの解 $\begin{cases} E_1(x) = -\nabla\phi_1 \\ E_2(x) = -\nabla\phi_2 \end{cases}$ $\qquad \nabla E_1 = \nabla E_2 = \frac{1}{\epsilon_0}\rho$

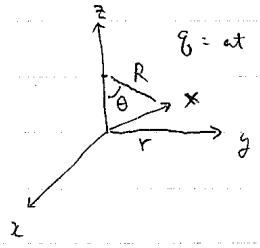
いま $\begin{cases} E = E_1 - E_2 \\ \phi = \phi_1 - \phi_2 \end{cases}$ $\rightarrow \nabla\cdot E = 0$ $\qquad \nabla\cdot(\phi E) = E\cdot\nabla\phi + \phi\nabla\cdot E = -E^2$

$$\int_V \nabla\cdot(\phi E)\,dV = \int_S \phi E\cdot d\mathbf{S}$$

$$-\int_V E^2\,dV \qquad \text{右辺}=0 \text{にする条件}$$

(1) $\phi = 0$ on S
電位を与える Dirichlet

(2) $E\cdot\hat{n} = 0$
電場の法線成分を与える Neumann.




$z=1$ at $q$ $\qquad \phi(x) = \frac{q}{4\pi\epsilon_0}\cdot\frac{1}{R}$

$$R = \sqrt{r^2+1-2r\mu} \qquad (\mu=\cos\theta)$$

$\phi$ を展開する $\quad \phi = \frac{q}{4\pi\epsilon_0}\sum_{n=0}^{\infty} r^n P_n(\mu)$

(6) で $A_2 \neq 0$ $\quad r\mu = z$

$$\phi = A_2 r^2 \frac{1}{2}(3\mu^2-1)$$

$$= \frac{1}{2}A_2(2z^2-x^2-y^2)$$

等ポテンシャル面 $x^2+y^2-2z^2 = \text{const}$

$M(p)/M(e^-) = 1836.152660\,(85)$
$M(\bar{p})/M(e^-) = 1836.152680\,(89)$

$$\begin{cases} \ddot{x} = -\omega_c(\dot{y} + \frac{A}{B}x) \\ \ddot{y} = \omega_c(\dot{x} - \frac{A}{B}y) \\ \ddot{z} = -2\frac{e}{m}Az \end{cases}$$

$$\omega_1 = -\frac{1}{2}\left(\omega_c \pm \sqrt{\omega_c^2 - 4\omega_c A/B}\right)$$

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$M(p)/M(e^-) = 1836.152660$ (83)

$M(\bar{p})/M(e^-) = 1836.152680$ (99)

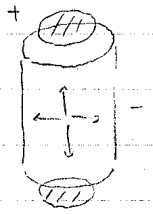
$e = 1.60 \times 10^{-19} C$

$m_p = 1.67 \times 10^{-27} kg$

$B \sim 1T$

$\frac{\omega_c}{2\pi} = 89 MHz$

$A \sim 10^4 V$




$$\ddot{x} = -\omega_c\left(\dot{y} + \frac{A}{B}x\right)$$

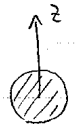
$$\ddot{y} = +\omega_c\left(\dot{x} - \frac{A}{B}y\right)$$

$$\ddot{z} = -2\frac{e}{m}Az$$

$$\omega_1 = \frac{1}{2}\left(\omega_c \pm \sqrt{\omega_c^2 - 4\omega_c A/B}\right)$$

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応用例




導体球

$\phi_0 = -E_0\hat{z} = -E_0 r\cos\theta$

$A_1 = -E_0$

$A_n = 0 \quad (n \neq 0, 1)$

$\phi(R,\theta) = \phi_S$

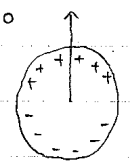
$\rightarrow B_1 = E_0 R^3$

$A_0 + B_0 R^{-1} = \phi_S$

$B_n = 0 \quad (n \neq 0, 1)$

$$\phi(r,\theta) = \phi_S + \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{r} - \frac{1}{R}\right) - E_0 r\left(1 - \frac{R^3}{r^3}\right)\cos\theta$$

$Q = 0$



分極 polarization

内部を遮へいする

表面電荷

$$\sigma_S = \epsilon_0 E_r\big|_{r=R} = \frac{Q}{4\pi R^2} + 3\epsilon_0 E_0\cos\theta$$

中心に dipole $P = 4\pi\epsilon_0 R^3 E_0$

体積に比例

$$E = \frac{1}{4\pi\epsilon_0}\cdot\frac{1}{r^3}\left[3(P\cdot X)X - P\right]$$

$\frac{1}{2}mv^2 + \frac{1}{2}\cdot\frac{4\pi R^3}{3}\rho\frac{v^2}{2}$

$m^* = m + \frac{1}{2}M$

(問題)

(1) 2次元 i.e. 導体円柱(軸 // x) $E_0 // \hat{z}$

(2) 導体球(半径 R)

・$\sigma$

・全電荷

・$R \to \infty$

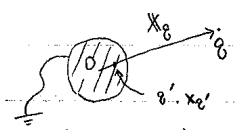
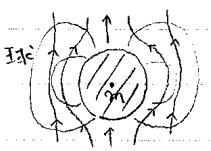



Image の方法

(Mirror image)

コメント 非圧縮性完全流体

$\nabla\cdot v = 0$

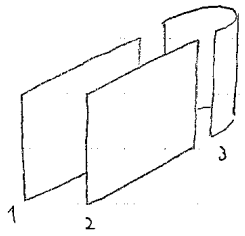



b.c. $v\cdot\hat{n} = \nabla\phi\cdot\hat{n} = 0$, $\frac{\partial\phi}{\partial r}\big|_{r=R} = 0$

$\rightarrow \phi_0 = v_0 r\left(1 + \frac{1}{2}\cdot\frac{R^3}{r^3}\right)\cos\theta$

速度ポテンシャル

## §5 静電容量 Capacitance




導体 1. 2. 3
電位 $V_1$ $V_2$ $V_3$
全電荷 $Q_1$ $Q_2$ $Q_3$

・電場の potential 一定 → 電位
・各導体の $V_i$ あるいは $Q_j$ を与えれば
$\phi(x)$ unique

$$\int \phi\,(E_{(1)}-E_{(2)})\cdot dS = V_i \int (E_{(1)}-E_{(2)})\cdot dS$$
$$= V_i \frac{1}{\epsilon_0}(Q_{i(1)}-Q_{i(2)}) \qquad V_i = V_{(1)i}-V_{(2)i}$$

電場のエネルギー

$$\mathcal{E} = \frac{1}{2}\int \rho(x)\phi(x)\,dV$$
$$= \frac{1}{2}\sum V_i Q_i$$

真空中 重ね合せ

いま $\begin{cases} V_i = 0 & i \neq j \\ V_j = 1 \end{cases}$ → $\phi^{(j)}(x)$ と書く

$$\phi(x) = \sum_j \phi^{(j)}(x)\,V_j$$

このとき、i番目の導体の電荷

$$Q_i^{(j)} = \int_{S_i} \sigma^{(j)}(x)\,dS = C_{ij}$$

$\phi^{(i)} = 1$ on $S_i$

$$Q_i^{(j)} = \int \phi^{(i)}(x)\,\sigma^{(j)}(x)\,dS = \frac{1}{4\pi\epsilon_0}\iint \frac{\sigma^{(i)}(x')\,\sigma^{(j)}(x)}{|x'-x|}\,dS\,dS'$$

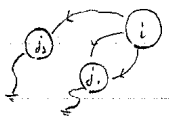
Coulomb の法則

$$\phi^{(i)}(x) = \frac{1}{4\pi\epsilon_0}\int \frac{\sigma^{(i)}(x)}{|x-x'|}\,dS'$$

すべての導体上 / すべての導体

・$C_{ij} = C_{ji}$
・$Q_i = C_{ij}V_j$
$$\mathcal{E} = \frac{1}{2}\sum C_{ij}V_iV_j$$

$C_{ii} \geq 0$
$C_{ij} \leq 0$ $(i \neq j)$




例 (a) 球 R

$C_{11} = C$

$$\phi = \frac{1}{4\pi\epsilon_0 r}Q$$
$$C = 4\pi\epsilon_0 R$$

(b)




$d \ll L$
面積 A

$$1 = Ed = \frac{Q}{\epsilon A}d$$
$$C_{12} = \epsilon_0\frac{A}{d}$$

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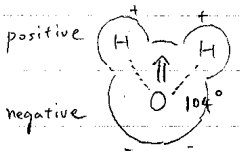
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# 第三章 誘電体

## §1

$10^{-8}$ cm → 平均 Conductor $\rho_{free}$ Insulators 束ばくされた電荷

Dielectrics (強誘電体) ferro electrics




$P = 0.62 \times 10^{-29}$ Coulomb · meter

$\mathbb{E}$ マクロの場 → $\mathbb{E}(\mathbb{x}) = \int \mathbb{E}_{micro}(\mathbb{x}')\, f(\mathbb{x}-\mathbb{x}')\, d\mathbb{x}'$




$\int f\, dV = 1$

$$\begin{cases} \nabla \cdot \mathbb{E}_{micro} = \frac{1}{\epsilon_0}\rho_{micro}(\mathbb{x}) \\ \nabla \times \mathbb{E}_m = 0 \end{cases}$$

$$\rightarrow \nabla \cdot \mathbb{E} = \int \mathbb{E}_m(\mathbb{x}') \cdot \nabla f(\mathbb{x}-\mathbb{x}')\, dV$$

$$= \int \nabla' \mathbb{E}_m\, f(\mathbb{x}-\mathbb{x}')$$

$$\downarrow$$

$$\int \frac{1}{\epsilon_0}\rho_m f\, dV'$$

$$\bar{\rho}$$

$\bar{\rho} = \rho_{free}(\mathbb{x}) - \nabla \cdot \mathbb{P}$

$\mathbb{P}(\mathbb{x})$ Polarization の場

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$\mathbb{D}(\mathbb{x}) = \epsilon_0 \mathbb{E} + \mathbb{P}$ $\rho(\mathbb{x}) = \int \rho_{micro}(\mathbb{x}')\, f(\mathbb{x}-\mathbb{x}')\, d\mathbb{V}'$

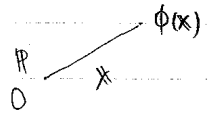
## §2. 分極 Polarization

個々の分子の dipole moment $\mathbb{P}_i$

分極密度 $\mathbb{P}(\mathbb{x}) = \sum_i \mathbb{P}_i\, \delta(\mathbb{x}-\mathbb{x}_i)$

$\mathbb{P}(\mathbb{x})\Delta V$ : $\Delta V$ のもつ d.m.

$$\phi(\mathbb{x}) = \frac{\mathbb{x} \cdot \mathbb{P}}{4\pi\epsilon_0 |\mathbb{x}|^3}$$




$$= \frac{1}{4\pi\epsilon_0} \nabla \frac{1}{|\mathbb{x}|} \cdot \mathbb{P}$$

$$\phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\int_{V'} \nabla' \frac{1}{|\mathbf{x}-\mathbf{x}'|}\mathbb{P}(\mathbf{x}')\,dV' = \frac{1}{4\pi\epsilon_0}\left\{\iint_{S'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\mathbb{P}(\mathbf{x}')\,dS' - \int_V \frac{1}{|\mathbf{x}-\mathbf{x}'|}\nabla'\cdot\mathbb{P}(\mathbf{x}')\,dV'\right.$$

誘電体の占める $V$

第一項　$\sigma_p(\mathbf{x}') = \hat{n}'\,\mathbb{P}(\mathbf{x}')$　表面

第二項　$-\nabla'\cdot\mathbb{P}(\mathbf{x}')$　電荷密度

$$\nabla\mathbb{E}_p = -\frac{1}{\epsilon_0}\nabla\mathbb{P}$$

free な電荷

+)　$\nabla\mathbb{E}_{free} = -\frac{1}{\epsilon_0}\int \rho_{free}$

$$\begin{cases} \mathbb{D}(\mathbf{x}) = \epsilon_0\mathbb{E}(\mathbf{x}) + \mathbb{P}(\mathbf{x}) \\ \nabla\cdot\mathbb{D} = \rho \\ \nabla\times\mathbb{E} = 0 \end{cases}$$

⊕ $\mathbb{E}$ と $\mathbb{P}$ or $\mathbb{E}$ と $\mathbb{D}$ の関係

$$\nabla\cdot\mathbb{E} = -\frac{1}{\epsilon_0}(\rho_{free} - \nabla\cdot\mathbb{P})$$

構成方程式

$$\begin{cases} \mathbb{P} = \epsilon_0\chi\mathbb{E} \\ \mathbb{D} = \epsilon_0 K\mathbb{E} = \epsilon\mathbb{E} \end{cases}$$

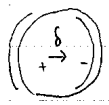
$\chi$：電気感応率　$K = 1+\chi$

$\epsilon$：誘電率

K：　1.00059　air

78.5　$H_2O$

Ze 電子



$$E \quad ZeE = \left(\frac{\delta}{R}\right)^3\frac{Ze\,Ze}{4\pi\epsilon_0\delta^2}$$

$$\delta = 4\pi\epsilon_0\frac{R^3}{Ze}E$$

双極子　$Ze\,\delta$　$d = 4\pi\epsilon_0 R^3$

$\chi = n\cdot 4\pi R^3$

## §3. 誘電体のあるときの電場

$$\mathbb{E} = -\nabla\phi$$

$$\mathbb{D} = \epsilon\mathbb{E}$$

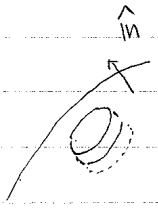
$$(\mathbb{E}_2 - \mathbb{E}_1)\cdot\hat{n} = \frac{1}{\epsilon_0}(\sigma_{free} + \sigma_p)$$

$$\longrightarrow \quad (\mathbb{D}_2 - \mathbb{D}_1)\cdot\hat{n} = \sigma_{free}$$

$$\nabla\times\mathbb{E} = 0$$

$$\longrightarrow \quad (\mathbb{E}_2 - \mathbb{E}_1)\cdot\hat{t} = 0$$

2.




境界面

$$\begin{cases} \sigma_{p1} = \hat{n}\,\mathbb{P}_1 \\ \sigma_{p2} = -\hat{n}\,\mathbb{P}_2 \end{cases}$$

↓

表面電荷密度

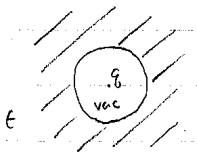
$$(\mathbb{P}_1 - \mathbb{P}_2)\hat{n} = \sigma_p$$

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Potential $\varphi_1(x) = \varphi_2(x)$

例1) Point charge $q$




Inside $D_1 = \frac{q}{4\pi r^2} = \epsilon_0 E_1$

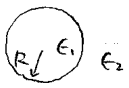
Outside $D_2 = \epsilon E_2$ $\quad D_1(R) = D_2(R)$

$$E_2(r) = \frac{q}{4\pi\epsilon r^2} \quad (r > R)$$

$r = R$; $E_2 - E_1 = \frac{1}{\epsilon_0}\sigma_p$

$$\rightarrow \sigma_p = \frac{q}{4\pi R^2}\left(\frac{\epsilon_0}{\epsilon} - 1\right)$$

(1) $\uparrow \mathbf{E} = E_0 \hat{z}$ $\quad \nabla \cdot \mathbf{D} = 0$ $\quad E_r = -\frac{\partial \phi}{\partial r}$



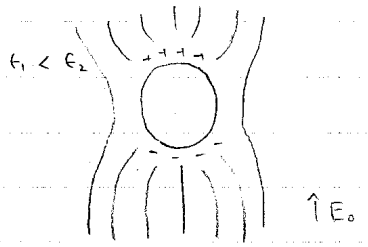

$\mathbf{D} = \epsilon \mathbf{E} = -\epsilon \nabla \phi$

$\rightarrow \nabla \phi^2 = 0$

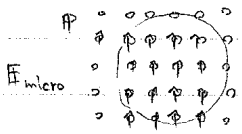
$$\epsilon_1 \left.\frac{\partial \phi_1}{\partial r}\right|_{r=R} = \epsilon_2 \left.\frac{\partial \phi_2}{\partial r}\right|_{r=R}$$

$$\phi_1 = -\frac{3\epsilon_2 E_0}{2\epsilon_2 + \epsilon_1} r\cos\theta$$

$$\phi_2 = -E_0 r\cos\theta + \frac{\epsilon_1 - \epsilon_2}{\epsilon_1 + 2\epsilon_2} E_0 R^3 \frac{\cos\theta}{r^2}$$




ミクロとマクロの関係




Clausius - Mossetti

$$\mathbf{P} = n\mathbf{p} = n\alpha \mathbf{E}_{micro} = n\alpha\left(\mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0}\right)$$

$$\mathbf{P} = \frac{n\alpha}{1 - \frac{n\alpha}{3\epsilon_0}} \mathbf{E}$$

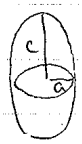
Jun 11

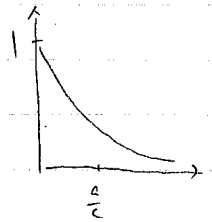
Ellipsoid $\mathbf{P}$ 一定

$\mathbf{E}_1 = \mathbf{E}_0 + \mathbf{E}_{dep}$

Depolarization field

$E_{dep,x} = -\frac{N_x}{\epsilon_0} P_x$, ... $y$ $z$ 成分



## §4 電場のエネルギー

$$\mathcal{E} = \frac{1}{2}\int \mathbb{D}\cdot\mathbb{E}\, d\overline{V}$$

$$\mathbb{D} = \epsilon\mathbb{E}$$

Conductor



誘電体（真空も含める）

導体上の

$\sigma = -\mathbb{D}\cdot\hat{n}$

$i$番目の Cond. の全電荷 $Q_i$

$$Q_i = -\int_{S_i}\mathbb{D}\cdot\hat{n}\, dS \qquad (1)$$

$Q_i \to Q_i + \delta Q_i$

表面上で $\phi_i$ = 一定

$$\delta\mathcal{E} = \sum \phi_i \delta Q_i \qquad (2)$$

$\phi_i(Q_1, Q_2, \cdots)$

$$\delta Q_i = -\int_{S_i}\delta\mathbb{D}\cdot\hat{n}\, dS \qquad (3)$$

(2), (3) →

$$\delta\mathcal{E} = -\sum_i\int_{S_i}\phi_i\,\delta\mathbb{D}\cdot\hat{n}\, dS \qquad (4)$$

$$\nabla\cdot(\phi\,\delta\mathbb{D}) = \nabla\phi\cdot\delta\mathbb{D} + \phi\underbrace{\nabla\cdot\delta\mathbb{D}}_{=0}$$

$$\boxed{\delta\mathcal{E} = \int \delta\mathbb{D}\cdot\mathbb{E}\, d\overline{V} \qquad (5)}$$

もし $\mathbb{D} = \epsilon\mathbb{E}$ → $\frac{1}{2}\epsilon\delta E^2$ $\qquad \mathcal{E} = \frac{1}{2}\int D\cdot E\, d\overline{V}$

(注意)

Cond の系 $\quad \mathcal{E} = \frac{1}{2}\sum\phi_i Q_i \quad \leftarrow \int\rho\phi\, d\overline{V}$ $\qquad Q_i = c_{ij}\phi_j$

$$\delta\mathcal{E} = \frac{1}{2}\sum(\phi_i\delta Q_i + \delta\phi_i Q_i)$$

ここで $\delta\phi_i$ は $\delta Q_i$ からくる変化．(2) と比べると $Q_i\delta\phi_i = \phi_i\delta Q_i$

力学的変化 も含むとき

熱力学的

$\underset{\sim}{\mathcal{E}}(Q_1, Q_2, \cdots, \xi_a, \xi_b, \cdots)$

電場のエネルギー

変化の仕方

1) $Q_i$ を fix して $\xi$ の変化 $\quad -\left.\frac{\partial\mathcal{E}}{\partial\xi}\right|_Q = F$

$F$ は外界に及ぼす力 $\quad F\delta\xi$

2) $\phi_i$ を fix $\quad \delta\xi$ にともなって $\delta Q$ も生じる．

$$\delta\mathcal{E} = -F\delta\xi + \phi\,\delta Q$$

一方 $\delta\mathcal{E} = \frac{1}{2}\phi\,\delta Q$ $\qquad \therefore \left.\frac{\partial\mathcal{E}}{\partial\xi}\right|_\phi = F$

もし $\bar{\mathcal{E}} \equiv \mathcal{E} - \phi Q$

$$\left.\frac{\partial\bar{\mathcal{E}}}{\partial\xi}\right|_\phi = F - \phi\frac{\partial Q}{\partial\xi} = -F$$

体積一定

$dQ = dE \qquad dE = dQ - p\,d\overline{V}$

圧力一定

$dQ = d\overline{W}$

$\overline{W}$：エンタルピー $\qquad W = E + p\overline{V}$

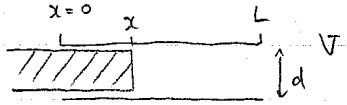
$C_V = \left.\frac{\partial E}{\partial T}\right|_V$

$C_p = \frac{\partial W}{\partial T}$

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Ex)




edge 効果は無視

V fix

$$\mathcal{E}(V, x) = \frac{1}{2}\cdot\frac{V^2}{d}\left[\varepsilon_0 L + (\varepsilon-\varepsilon_0)x\right] \quad \cdots ①$$

Q fix

$$\mathcal{E}(Q, x) = \frac{1}{2}Q^2 d\cdot\frac{1}{\varepsilon_0\left[L + \left(\frac{\varepsilon}{\varepsilon_0}-1\right)x\right]} \quad \cdots ②$$

① → $F = \left.\frac{\partial \mathcal{E}}{\partial x}\right|_V = \frac{1}{2}\frac{V^2}{d}(\varepsilon-\varepsilon_0)$

② → $F = -\left.\frac{\partial \mathcal{E}}{\partial x}\right|_Q = \frac{1}{2}\frac{V^2}{d}(\varepsilon-\varepsilon_0)$

$\left.\frac{\partial \mathcal{E}}{\partial \varepsilon}\right|_{q_i} = -F$

$\left.\frac{\partial \mathcal{E}}{\partial \varepsilon}\right|_{\phi} = F$

# 第4章 静磁場

$\nabla \times \mathbb{B} = \mu_0 \mathbb{J}$ (1)　　$\mathbb{B}$ 磁束密度　Tesla $= 10^4$ Gauss

$\nabla \cdot \mathbb{B} = 0$

$\mathbb{J}$ 定常電流　　$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbb{J} = 0$

角運動量 スピン　永久電流　　$\rho \to E \to \phi$

$\mathbb{J} \to \mathbb{B} \to \mathbb{A}$

vector potential

$\nabla \cdot \mathbb{B} = 0$ より $\boxed{\mathbb{B}(\mathbb{x}) = \nabla \times \mathbb{A}(\mathbb{x})}$ → $\nabla \cdot \mathbb{B} = 0$ identically

しかし $\mathbb{A}$ は unique ではない　$\mathbb{A} \to \mathbb{A} + \nabla \chi$

gauge 変換 の一つ

$\chi(\mathbb{x})$ スカラー場　$\nabla \times \nabla \chi(\mathbb{x}) = 0$

(1) → $\nabla \times (\nabla \times \mathbb{A}) = \mu_0 \mathbb{J}$　　gauge のとり方で $\boxed{\nabla \cdot \mathbb{A} = 0}$ とすることができる

$\nabla \cdot (\nabla \cdot \mathbb{A}) - \nabla^2 \mathbb{A} = \mu_0 \mathbb{J}$

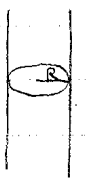
→ $\boxed{-\nabla^2 \mathbb{A} = \mu_0 \mathbb{J}}$　3つの方程式

$$\boxed{\mathbb{A}(\mathbb{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbb{J}(\mathbb{x}')}{|\mathbb{x} - \mathbb{x}'|} d\mathbb{V}'} \quad \to \quad \mathbb{B}(\mathbb{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbb{J}(\mathbb{x}') \times (\mathbb{x} - \mathbb{x}')}{|\mathbb{x} - \mathbb{x}'|^3} d\mathbb{V}'$$

線電流　$\mathbb{A}(\mathbb{x}) = \frac{\mu_0}{4\pi} \int \frac{d\mathbb{I}(\mathbb{x}')}{|\mathbb{x} - \mathbb{x}'|}$

Jun 25!

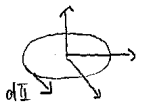
問題




一様な電流密度 $\mathbb{J}$

$\mathbb{B}(\mathbb{x})$, $\mathbb{A}(\mathbb{x})$

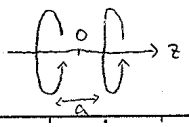
円型電流




$d\mathbb{I} = I\, d\mathbb{x}'$

$A_z = A_\rho = 0$　　$\mathbb{m} = \pi R^2 I \hat{z}$ を原点においた時の場

問題 Helmholtz coil




Show that $\frac{dB}{dz} = \frac{d^2B}{dz^2} = 0$

$B(0)$ ?

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## §4.3 局在した $\mathbf{J}$

遠方で $\mathbf{J}=0$
定常 $\nabla\cdot\mathbf{J}=0$




$\int (\mathbf{x}\cdot\mathbf{x}')\,\mathbf{J}(\mathbf{x}')\,d\mathbf{V}'$

磁気双極子 $\mathbf{m}\times\mathbf{x}$

Jackson $\mathbf{J}\to\rho(\mathbf{x}')$

$\mathbf{d}\to\int\mathbf{x}'\rho(\mathbf{x}')\,d\mathbf{V}'$ $\qquad \mathbf{J}=\sum q_\alpha \mathbf{v}_\alpha \delta(\mathbf{x}-\mathbf{x}_\alpha)$

$\phi(x)\propto\frac{\mathbf{d}\cdot\mathbf{x}}{|\mathbf{x}|^3}$ $\qquad \mathbf{m}=\frac{1}{2}\sum_\alpha q_\alpha \mathbf{L}_\alpha/m_\alpha$ $\qquad$ Bohr magneton $\frac{e\hbar}{2m}$

$\mathbf{m}=-\frac{|e|}{2m}\mathbf{L}$

$\mathbf{B}(\mathbf{x})=$ (5)式 $+\frac{\mu_0}{4\pi}\cdot\frac{8\pi}{3}\mathbf{m}\,\delta(\mathbf{x})$

原点 $\mathbf{x}=0$ に $\mathbf{m}$ があるとき

## §4.4 磁気的な力 トルク

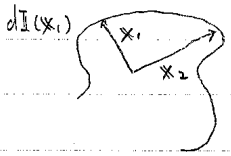
静磁場中で受ける力

出発点 Lorentz力 $\mathbf{F}=q\mathbf{E}+q\mathbf{v}\times\mathbf{B}(\mathbf{x})$

$\Delta V$ の中 $\Delta\mathbf{F}=\mathbf{J}(\mathbf{x})\times\mathbf{B}(\mathbf{x})\,\Delta V$

$$\boxed{\mathbf{F}=\int\mathbf{J}(\mathbf{x})\times\mathbf{B}(\mathbf{x})\,dV} \qquad (1)$$

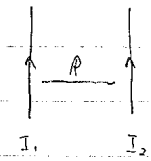
Ex) 線電流の間に働く力




$$d\mathbf{F}_{21}=d\mathbf{I}(\mathbf{x}_2)\cdot\frac{\mu_0}{4\pi}\cdot\frac{d\mathbf{I}(\mathbf{x}_1)\times(\mathbf{x}_2-\mathbf{x}_1)}{|\mathbf{x}_2-\mathbf{x}_1|^3} \qquad (2)$$

$$\mathbf{F}_{12}=\frac{\mu_0}{4\pi}I_1I_2\oint_{C_1}\oint_{C_2}\frac{d\mathbf{x}_1\times(d\mathbf{x}_2\times\mathbf{R})}{R^3} \qquad (3)$$ Ampére

2本の平行電流




$\mathbf{B}(\mathbf{x})=\frac{\mu_0}{2\pi\rho}I\hat{\varphi}$ $\qquad I_1=I_2=1\,A$

$\mathbf{f}_{12}=-\frac{\mu_0}{2\pi\rho}I_1I_2\hat{\rho}$ $\qquad \rho=1\,m$

単位長さあたり $\qquad f=2\times10^{-7}\ N/m$

局在した $\mathbf{J}$ に働く力

$\mathbf{B}$ 外場

$\mathbf{B}(\mathbf{x})\simeq\mathbf{B}(0)+(\mathbf{x}\cdot\nabla)\mathbf{B}\big|_{x=0}$ $\qquad \mathbf{F}=\int\mathbf{J}(\mathbf{x})\times(\mathbf{x}\cdot\nabla)\mathbf{B}\,dV$

$F_i=\epsilon_{ijk}\int J_j x_l\partial_l B_k\,dV$ $\qquad \boxed{\mathbf{F}=\nabla(\mathbf{m}\cdot\mathbf{B})}$ (5)

対応するエネルギー $\boxed{U_m = -m \cdot B}$ (6)

B ↑ m　一様なBから力は受けない，トルクは受ける．

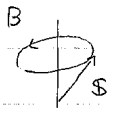
m ↑ δθ → m　δθ だけ回転　$m = m + \delta\theta\, m \times \hat{n}$

$\delta U_m = \delta\theta\, \hat{n}(m \times B)$

トルク N が働く　$\boxed{N = m \times B}$ (7)

NMR, ESR

スピン $ に伴なう $m = \gamma \$$　$\frac{d\$}{dt} = \gamma \$ \times B$




$\omega_0 \cong \frac{e}{2m_e} B$

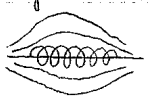
$e^-$ ; $\gamma = -\frac{e}{me}$

$10^{10}\ sec^{-1}$, B (Tesla)

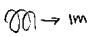
Jul. 2

応用例

Magnetic Mirror



$(m \cdot \nabla) B$



No. 16

Date 93・7・2

# 第5章 磁性体 magnetic materials

## §5-1 磁化 magnetization

分極 $\mathbb{P}$　　$\mathbb{M}(\mathbb{x})$ 磁気モーメントの密度　　$\mathbb{J}(\mathbb{x})$ と $\mathbb{M}(\mathbb{x})$ → $\mathbb{B}$

$$\mathbb{M}(\mathbb{x}) = \sum_\alpha n_\alpha(\mathbb{x}) \langle m_\alpha \rangle$$

$$\mathbb{A}(\mathbb{x}) = \frac{\mu_0}{4\pi}\left\{\int \frac{\mathbb{J}(\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|}\,d\mathbb{V}' + \int \frac{\mathbb{M}(\mathbb{x}')\times(\mathbb{x}-\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|^3}\,d\mathbb{V}'\right\} \qquad (1)$$

"

$$\int \mathbb{M}(\mathbb{x}')\times\nabla'\frac{1}{|\mathbb{x}-\mathbb{x}'|}\,d\mathbb{V}' = \int \frac{\nabla'\times\mathbb{M}(\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|} - \nabla'\times\left(\frac{\mathbb{M}(\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|}\right)d\mathbb{V}'$$

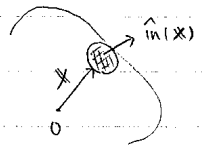
Gauss の定理の系

$\mathbb{C}\times\mathbb{A}$ に G の定理を適用

$$= \int_{S'} \frac{\mathbb{M}(\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|}\times d\mathbb{S}$$

$$\mathbb{A} = \frac{\mu_0}{4\pi}\left\{\int_V \frac{\mathbb{J}(\mathbb{x}')+\mathbb{J}_m(\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|}\,d\mathbb{V}' + \int_S \frac{\mathring{\mathbb{J}}_m(\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|}\,d\mathbb{S}'\right\} \qquad (4)$$

$$\left.\begin{aligned}\mathbb{J}_m(\mathbb{x}) &= \nabla\times\mathbb{M}(\mathbb{x})\\ \mathring{\mathbb{J}}_m(\mathbb{x}) &= \mathbb{M}(\mathbb{x})\times\hat{\mathbb{n}}(\mathbb{x})\end{aligned}\right\}(3) \qquad \nabla\cdot\mathbb{J}_m = 0$$




$\mathbb{M}$ があるとき

$\mathbb{H}(\mathbb{x})$ 磁場の強さ (magnetic intensity)

を導入

$$\nabla\times\mathbb{B} = \mu_0(\mathbb{J}+\mathbb{J}_m)$$

$$\nabla\times(\mathbb{B}-\mu_0\mathbb{M}) = \mu_0\mathbb{J}$$

$$\mathbb{H} = \mathbb{B}/\mu_0 - \mathbb{M} \qquad (5)$$

$$\left.\begin{aligned}\nabla\times\mathbb{H} &= \mathbb{J}\\ \nabla\cdot\mathbb{B} &= 0\end{aligned}\right\}(6)$$

対応する式

$$\left\{\begin{aligned}\mathbb{D} &= \epsilon_0\mathbb{E}+\mathbb{P}\\ \nabla\cdot\mathbb{D} &= \rho\\ \nabla\times\mathbb{E} &= 0\end{aligned}\right.$$

$\mathbb{M} = \mathbb{M}(\mathbb{H})$

Linear　$\mathbb{M} = \chi\mathbb{H}$　　$(M_i = \chi_{ij}H_j)$ (7)

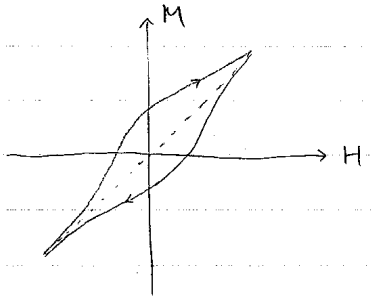
帯磁率 magnetic susceptibility　　$\chi > 0$ 常磁性　　$\mathbb{M}\neq 0$, $\mathbb{H}=0$ ; 強磁性

$\chi < 0$ 反 "

$\mathbb{B} = \underbrace{\mu_0(1+\chi)}_{\mu}\mathbb{H}$ (8)

$\mu$ 透磁率 permeability

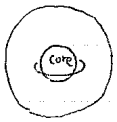
$\frac{\mu}{\mu_0}(1+\chi) = K_m$




moment $[Am^2]$

$\mu_B$ Bohr magneton
$= 9\times10^{-24}$

地球 $\sim 8.2\times10^{22}$




$3\times10^6$m
$I \sim 4\times10^{10}$A

磁石
Fe 数$\mu_B$
$n \sim 10^{23}/cm^3 = 10^{29}/m^3$
$|M| \sim 10^6\ A\cdot m^{-1}$

$B \sim \mu_0 M \sim 4\pi\ 10^{-7}\cdot 10^6 \sim 1$ Tesla



Best magnet 6 T

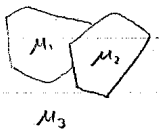


$j_m \sim 10^6\ A\,m^{-1}$

## §2. 境界値問題

(i) ベクトル potential

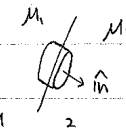
$\mathbb{B} = \mu\mathbb{H}$ が成り立つ




$\nabla\times(\nabla\times A) = \mu\mathbb{J}$

ゲージ $\nabla\cdot A = 0$

$-\nabla^2 A = \mu\mathbb{J}$ (1)

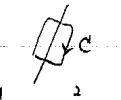



a) $\nabla\cdot\mathbb{B} = 0$

$(\mathbb{B}_2 - \mathbb{B}_1)\,\hat{n} = 0$ (2)

b) Stokes の定理

$\oint_C \mathbb{H}\,d\mathbb{l} = \oint \mathbb{J}\,d\mathbb{S}$




$\hat{n}\times(\mathbb{H}_2 - \mathbb{H}_1) = \mathbb{j}$ (3)

$\mathbb{j}$ は境界電流

(ii) 磁気ポテンシャル

$\mathbb{M}$ にともなう磁場

§1.(4) $\mathbb{B} = \frac{\mu_0}{4\pi}\nabla\times\int\frac{\mathbb{M}'\times(\mathbb{x}-\mathbb{x}')}{|\mathbb{x}-\mathbb{x}'|^3}d\mathbb{V}' = \frac{\mu_0}{4\pi}\nabla\times\int\mathbb{M}'\times\left(\nabla'\frac{1}{|\mathbb{x}-\mathbb{x}'|}\right)d\mathbb{V}$

$= \frac{\mu_0}{4\pi}\int\left[4\pi\mathbb{M}'\delta(\mathbb{x}-\mathbb{x}') - \nabla\left(\mathbb{M}'\cdot\nabla\frac{1}{|\mathbb{x}-\mathbb{x}'|}\right)\right]d\mathbb{V}'$

$\mathbb{B}(\mathbb{x}) = \mu_0\,\mathbb{M}(\mathbb{x}) + \mu_0\nabla\Phi_m(\mathbb{x})$

$\left(\Phi_m = \frac{1}{4\pi}\int\mathbb{M}'\nabla'\frac{1}{|\mathbb{x}-\mathbb{x}'|}d\mathbb{V}'\right)$ (1)

§1.(5) $\mathbb{H} = -\nabla\Phi_m$ (2)

No. 18

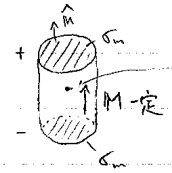
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$$\Phi_m = \frac{1}{4\pi}\int \sigma_m(x')\frac{1}{|x-x'|}dS' + \frac{1}{4\pi}\int \rho_m(x')\frac{1}{|x-x'|}dV' \qquad (3)$$

ここで

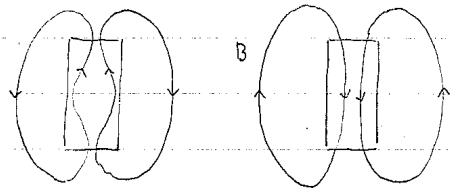
$$\rho_m = -\nabla M$$

$$\sigma_m = M\cdot\hat{n} \qquad (4)$$




中点 $B = \mu_0 M \frac{L}{\sqrt{L^2+R^2}}$

$$H = -M\left(1 - \frac{L}{\sqrt{L^2+R^2}}\right)$$




$J=0$ の領域

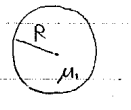
$\nabla\times H = 0$

$H = -\nabla\Phi_m$

もし $B=\mu H$, $\nabla\cdot B = 0$ → $\nabla(\mu\nabla\Phi_m) = 0$

さらに $\mu$ が piece wise $\nabla^2\Phi_m = 0$

例. 球




$H_0 = H_0\hat{z}$

# 第6章 電磁誘導

Maxwell

$\nabla\cdot E = \frac{1}{\epsilon_0}\rho$ $\quad \nabla\times E = -\frac{\partial B}{\partial t}$ 第1歩 準静的 quasi-static

$\nabla\cdot B = 0$ $\quad \nabla\times B = \mu_0 J + \mu_0\epsilon_0\frac{\partial E}{\partial t}$ $\quad \mu_0\epsilon_0 = \frac{1}{c^2}$

変化のスケール

空間 $\lambda$ $\quad \lambda^{-1}E \sim \omega B$

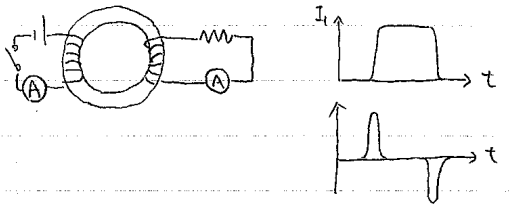
時間 $\omega$ $\quad \lambda^{-1}B = \mu J + \frac{1}{c^2}\omega E$

$\lambda^{-2}E \sim \frac{1}{c^2}\omega^2 E$

$\lambda^{-2} \gg \frac{\omega^2}{c^2} \rightarrow \frac{c^2}{\lambda^2} \gg \omega^2$

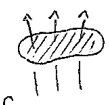
Faraday 電流 → B

1831 Henry B → 電気的な効果




$\nabla\times E + \frac{\partial B}{\partial t} = 0$ (1)

積分形



C 空間に固定

$$\oint_C E(x,t)\,dx + \int_S \frac{\partial B}{\partial t}\cdot dS$$

$$= \frac{d}{dt}\int_S B(x,t)\cdot dS(x)$$

磁束 flux

$V_S$ 2" const ($\because \nabla\cdot B = 0$)

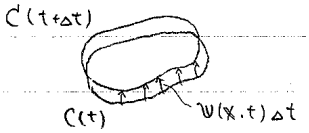
$$\Phi_c(t) = \int_{S_c} B\cdot dS \quad (2)$$

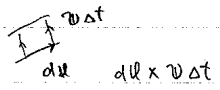
$$\mathcal{E}_c(t) = -\frac{d}{dt}\Phi_c(t) \quad (3)$$

$$\mathcal{E}_c(t) = \oint E(x,t)\,dx \quad (4)$$

C に そう 起電力

C が時間的に変化するとき




$$\Phi_{C(t+\Delta t)} = \int_{S(t+\Delta t)} B\cdot dS$$

$$\Phi_{C(t)} = \int_{S(t)} B\cdot dS$$

側面 $= \Delta t\int_C B\cdot(dl\times v)$

$= \Delta t\int_C (v(x,t)\times B(x,t))\,dx$

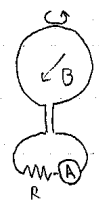
$$\mathcal{E}_C = -\frac{d\Phi_C^{(B)}}{dt} - \frac{d\Phi_C^{(C)}}{dt}$$

$$\Delta\Phi_C = \Delta t\int_{S(t)}\frac{\partial B}{\partial t}dS + \underbrace{\int_{S(t+\Delta t)} B(x,t)\,dS - \int_S \text{〃} + \text{側面}}_{\text{〃}\ 0\ \text{〃}}$$

0 $\quad -v\,0'$

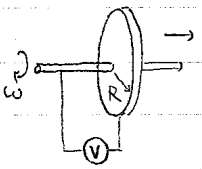
$E' = E + v\times B$

例1




π 回転

$\int_0^\infty I(t)\,dt$




$V = \frac{1}{2}\omega BR^2$

$R = 0.1\,m$ $\quad \frac{\omega}{2\pi} = 10^3\,Hz$ $\quad B = 0.1\,Tesla$

$V \sim 3\,volt$

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## §2. 磁場のエネルギー

$\nabla(E\times B) = E(\nabla\times B) - B(\nabla\times E)$

真空中　磁場の変化 $E$ → 仕事 $\delta W = E\cdot E = J\cdot E$

$$P(t) = \int J\cdot E\, d\tau \quad (1)$$

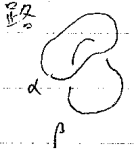
$J = \frac{1}{\mu_0}\nabla\times B\left(-\epsilon_0\frac{\partial E}{\partial t}\right)$ を代入　$P = -\frac{d}{dt}\int\frac{B^2}{2\mu_0}d\tau$ $\left(-\frac{d}{dt}\int\frac{\epsilon_0 E^2}{2}d\tau\right)$ $(E\times B = 0)$ (2)

$B = \nabla\times A$ ∴ 磁場のエネルギー $\boxed{U_B(t) = \int\frac{B^2}{2\mu_0}d\tau}$ (3)

$$U_B = \frac{1}{2}\int J\cdot A\, d\tau$$

$$= \frac{\mu_0}{4\pi}\iint\frac{J(x,t)\,J(x',t)}{|x-x'|}d\tau\, d\tau'$$

回路




$I_\alpha$ $I_\beta$

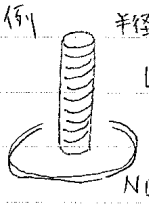
$$U_B = \frac{1}{2}\sum_{\alpha,\beta}L_{\alpha\beta}I_\alpha I_\beta \quad (6)$$

Inductance $L_{\alpha\beta} = \frac{\mu_0}{4\pi}\oint_\alpha\oint_\beta\frac{dx_\alpha\, dx_\beta}{|x_\alpha - x_\beta|}dx_\alpha\, dx_\beta$ (7)

$\Phi_c = \int_{S_c} B\cdot dS$ ($\nabla\times A$) $\boxed{\Phi_c = \oint_c A\cdot dx}$ (8)

$$U_B = \frac{1}{2}\sum_\alpha I_\alpha\Phi_\alpha \quad (9) \qquad L_{\alpha\beta} = \Phi_{\alpha\beta} \quad \beta\text{に}I_\beta = 1$$

例




半径 $a$　$n$回

L（単位長さあたり）$= \pi a^2 n^2\mu_0$

インダクタンス [Joule/A²]

$M = \pi a^2 nN\mu_0$

磁性体のあるとき

$B = \mu H$ が成り立つとき $\boxed{U_B = \frac{1}{2}\int H\cdot B\, d\tau}$ (9)　$\frac{1}{2}\int\epsilon E^2 d\tau$ $\epsilon > \epsilon_0$

$\frac{1}{2m}\left(P - \frac{e}{c}A\right)^2$　$= \frac{1}{2}\int\frac{B^2}{\mu}d\tau$

$\mu > 0$　$\mu > \mu_0$ 常磁性

$<$ 反磁性

# 第7章 単位系

{静電単位
電磁 〃

J.C. Maxwell & F. Jenkins 1862

物理

基本的次元 力学

長さ[L] 時間[T] 質量[M]

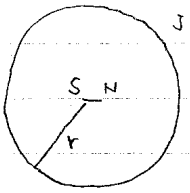
導かれる次元 力 $[ML/T^2]$

電磁気

1) 磁気 (電流) 1) $f = \frac{m_1 m_2}{r^2}$ 磁荷 m $[M^{1/2} L^{3/2} T^{-1}]$

2) 電気 (電荷) ↓力学 モーメント $\mu = m\ell$ $[M^{1/2} L^{5/2} T^{-1}]$

電流

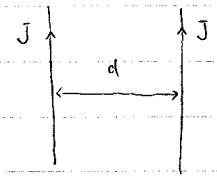



a) 磁石への作用

トルク $= \mu J 2\pi r \frac{1}{r^2} \cos\theta$

$\frac{ML^2}{T^2} = \frac{M^{1/2} L^{5/2}}{T} \cdot [J] \cdot \frac{1}{L}$ $\quad [J] = \frac{M^{1/2} L^{1/2}}{T}$

b) 電流間の力




単位長さあたりの力 $= 2C_m J^2/d$

$\frac{ML}{T^2} \cdot \frac{1}{L} = [J]^2/L$ $\quad [J] = \left[\frac{ML}{T^2}\right]^{1/2}$

$C_m = \mu_0/4\pi = 10^{-7}$

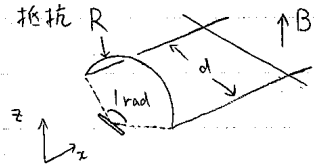
$d = 1\,m$ $10^{-7}\,N/m$ $\mu_0 = [N/A^2]$

1 Ampére · 1 sec $\longrightarrow$ $Q_m = [M^{1/2} L^{1/2}]$ Coulomb

起電力 $V \cdot Q = W$

$V = W/JT$

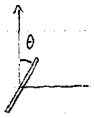
$R = V/J = W/J^2T = [L/T]$ 速度の次元をもつ




$V = v d B_0 \longrightarrow I = v d B_0/R$

→ 原点に作る磁場 $B_x = \frac{\mu_0}{4\pi} \frac{I}{d}$

$\tan\theta = \frac{\mu_0}{4\pi} \frac{v}{R}$




$\theta = 45°$ $\quad R = \frac{\mu_0}{4\pi} v$

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静電単位

Coulomb $f = \frac{qq'}{r^2}$ $\quad Q_e = \left[\frac{M^{1/2}L^{3/2}}{T}\right]$ $\quad$ cf. $Q_m = [M^{1/2}L^{1/2}]$

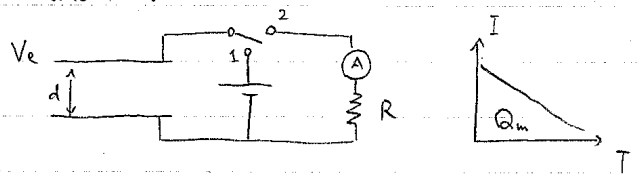
$[Q_e]/[Q_m] = L/T = c$ Weber · Kohlrausch 1856

$[J]_e = M^{1/2}L^{3/2}T^{-2}$

$[V]_e = M^{1/2}L^{1/2}T^{-1}$

$[R]_e = TL^{-1}$

a)




$C_e = L^2/4\pi d$ $\quad Q_e = C_e V_e$

$$\begin{cases} \nabla\cdot E = \frac{\rho}{\epsilon} \\ \nabla\times E = -\frac{\partial B}{\partial t} \\ \nabla\times B = \mu_0 J + \mu_0\epsilon_0\frac{\partial E}{\partial t} \\ \nabla\cdot B = 0 \end{cases}$$

$B \to \frac{1}{c}B'$

$\nabla\times E = -\frac{1}{c}\frac{\partial B}{\partial t}$

$\nabla\times B = c\mu_0 J + c\mu_0\epsilon_0\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 c}J + \frac{1}{c}\frac{\partial E}{\partial t}$

$F = \rho E + J\times B$ $\qquad F = \rho E + \frac{1}{c}J\times B'$

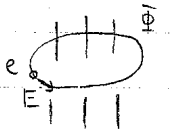
$\epsilon_0 = \frac{1}{4\pi}$

$\nabla\cdot E = 4\pi\rho$

$\nabla\times E = -\frac{1}{c}\frac{\partial B'}{\partial t}$

$\nabla\times B = \frac{4\pi}{c}J + \frac{1}{c}\frac{\partial E}{\partial t}$

補足) ベクトルポテンシャル




$\mathcal{E} = -\frac{d\Phi}{dt} = 2\pi r E$ $\qquad E = -\frac{d\Phi}{dt}\cdot\frac{1}{2\pi r}$

$\dot{p} = eE = \frac{e}{2\pi r}\cdot\frac{d\Phi}{dt} \to p = \frac{e}{2\pi r}\Phi = \frac{2\pi\hbar}{\lambda} = \frac{N\hbar}{r}$ $\qquad N\lambda = 2\pi r$

$\Phi = \frac{N(2\pi)\hbar}{e} = N\left(\frac{h}{e}\right)$

# 第9章 定常電流

## EMF 起電力

### §1 電流

a) 電荷の移動 ——— { Convection current, 水滴, 物体の運動

b) 変位電流 → 非定常のときだけ { Conduction current, マクロには中性, ⊕ ⊖ の相対運動

$10^{22}$ electrons / $cm^3$

$d\vec{a}$ $dI = \sum_\alpha \rho_\alpha \langle \vec{v}_\alpha \rangle \cdot d\vec{a}$

↖ drift velocity

$10^{-6}\,m^2$ → 1A

$|\langle \vec{v} \rangle| \sim 6 \times 10^{-4}$ m/s

$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$

$\vec{J}(\vec{x}) = \sigma_c \vec{E}(\vec{x})$

定常 $\nabla \cdot \vec{J} = 0$

$\sigma_c$ : conductivity 非等方性 → テンソル

オームの法則 $V = RI$.

$\rho : \sigma_c^{-1}$ resistivity

R : Resistance $R^{-1}$ : Conductance

磁場 Hall効果

### §2 定常電流分布

$\nabla \cdot \vec{J} = 0$

$(\vec{B} = 0)$

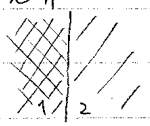
$\nabla \times \vec{E} = 0 \longrightarrow \nabla \times \sigma_c^{-1} \vec{J} = 0$

$\nabla \cdot \vec{E} = \frac{1}{\varepsilon_0}\rho \qquad \nabla \cdot \sigma_c^{-1} \vec{J} = \frac{1}{\varepsilon_0}\rho$

Joule 熱

$P = \int \vec{J} \cdot \vec{E}\, dV = \int_V \sigma_c |\vec{E}|^2 dV$

境界

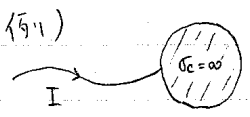



$\vec{J}_1 \cdot \vec{n} = \vec{J}_2 \cdot \vec{n}_2$

$\vec{E}_{1t} = \vec{E}_{2t} \rightarrow \frac{\vec{J}_{1t}}{\sigma_{c1}} = \frac{\vec{J}_{2t}}{\sigma_{c2}}$

表面電荷 $\sigma = J_n\left(\frac{\varepsilon_2}{\sigma_{c2}} - \frac{\varepsilon_1}{\sigma_{c1}}\right)$

例1) 導体球 $r_0$




$\sigma_c$ $\varepsilon_0$

$\vec{J}(r) = \frac{I}{4\pi r^2}\hat{r}$

$\vec{E}(r) = \frac{I}{4\pi\sigma_c r^2}\hat{r}$

$V_\infty = 0$ とする

球の電位 $V = -\int_\infty^{r_0} \vec{E}\, d\vec{r} = \frac{I}{4\pi\sigma_c r_0}$

$R = \frac{V}{I} = \frac{1}{4\pi\sigma_c r_0}$

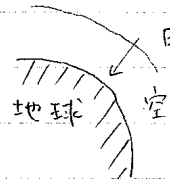
表面電荷 $\sigma = \frac{\varepsilon_0 I}{4\pi \sigma_c r_0^2} \rightarrow Q = \frac{\varepsilon_0}{\sigma_c} I$

Capacitance $C = \frac{Q}{V} = 4\pi\varepsilon_0 r_0$ $\therefore RC = \frac{\varepsilon_0}{\sigma_c}$

(電線+)

Leak 始めに $Q_0$ $\frac{dQ}{dt} = -I(t) = -\frac{\sigma_c}{\varepsilon_0} Q \rightarrow Q = Q_0 e^{-\frac{t}{RC}}$

$\left(\tau = \frac{\varepsilon_0}{\sigma_c}\right)$




$\sigma_c = (3.0 + 0.5 z^2) \times 10^{-14}\ (\Omega m)^{-1}$

$z$：高さ (km)

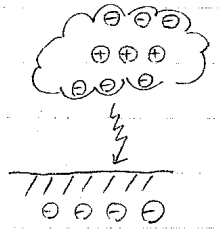
$\tau \sim \frac{10^{-11}}{10^{-14}} = 10^3\ s \sim 5$ 分

$\sigma_c$ (earth) $\sim 10^{-4}\ (\Omega m)^{-1}$

$J \sim 3 \times 10^{-12}\ A/m^2$

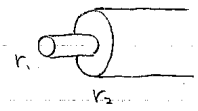
$\sigma \sim -10^{-19}\ C/m^2$

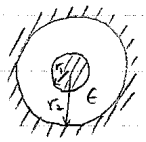
$Q \sim -5 \times 10^5\ C$



Copper ; $\sigma \sim 6 \times 10^7\ (\Omega m)^{-1}$

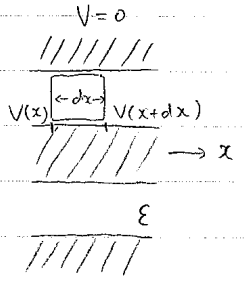
同軸ケーブル




単位長さ $L = \frac{\mu_0}{2\pi} \ln \frac{r_2}{r_1}$

$C = \frac{2\pi\varepsilon}{\ln \frac{r_2}{r_1}}$




$\frac{\partial V}{\partial x} dx = -L \frac{\partial I}{\partial t} dx - RI\, dx$

Leak なし

$\rightarrow \frac{\partial Q}{\partial t} = -\frac{\partial I}{\partial x}$, $Q = CV$ $\Big\} \quad C \frac{\partial V}{\partial t} = -\frac{\partial I}{\partial x}$

$\rightarrow LC \frac{\partial^2 I}{\partial t^2} + RC \frac{\partial I}{\partial t} = \frac{\partial^2 I}{\partial x^2}$ (1855)

(cut)

$LC = \varepsilon \mu_0 = \frac{\varepsilon}{\varepsilon_0 c^2}$

$R = \frac{1}{\pi r_1^2 \sigma_c}$

$\sigma_c \sim 6 \times 10^7$

$r_1 \sim 10^{-2}\ m$

$RC = \frac{1}{D} = \frac{1}{3 \times 10^{14}}$

Diffusion $\frac{\partial I}{\partial t} = D \frac{\partial^2 I}{\partial x^2}$ $T \sim \frac{L^2}{D}$

Leak があると $+\frac{1}{\tau_p} I$

$\sim \tau_p = C r_p$

# 第9章 時間的に変化する場

準定常

$\frac{\partial \rho}{\partial t} + \nabla \cdot J = 0$ (5)

Coulomb $\nabla \cdot E = \frac{1}{\varepsilon_0}\rho$ (1)

Ampére $\nabla \times B = \mu_0 J$ (2) → $\nabla \cdot$ (2) ; $\nabla \cdot J = 0$ 矛盾を含む

Faraday $\nabla \times E + \frac{\partial B}{\partial t} = 0$ (3)

$\nabla \cdot B = 0$ (4)

$\frac{\partial}{\partial t}$(1) ; $\nabla \cdot \frac{\partial E}{\partial t} = \frac{1}{\varepsilon_0}\frac{\partial \rho}{\partial t}$ --- (1)'

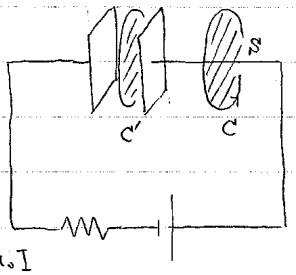
(1)' × $\mu_0\varepsilon_0$ + $\nabla \cdot$ (2) ; $\nabla \cdot \left\{ \nabla \times B + \varepsilon_0\mu_0 \frac{\partial E}{\partial t} \right\} = \mu_0 \left( \nabla \cdot J + \frac{\partial \rho}{\partial t} \right) = 0$ ☺ (5)

もともと (2) が

$\frac{1}{\mu_0} \nabla \times B = J + \varepsilon_0 \frac{\partial E}{\partial t}$ (6)

$E = \frac{Q}{\varepsilon_0 a}$

$\frac{dE}{dt} = \frac{1}{\varepsilon_0 a} I$




$\varepsilon_0\mu_0 \int \frac{dE}{dt} da = \mu_0 I$

$\int_C B \cdot dl = \mu_0 I$

(CGS) $\nabla \cdot E = 4\pi\rho$

$\nabla \times B - \frac{1}{c}\frac{\partial E}{\partial t} = \frac{4\pi}{c} J$

$\nabla \times E + \frac{1}{c}\frac{\partial B}{\partial t} = 0$

$\nabla \cdot B = 0$ covariant form

## §2 Potential による表示 Gauge

static $\nabla \times E = 0$ → scalar potential

t による $\neq 0$

$\nabla \cdot B = 0$ → vector potential $B(x,t) = \nabla \times A(x,t)$

(3) → $\nabla \times \left( E + \frac{\partial A}{\partial t} \right) = 0$ → $E + \frac{\partial A}{\partial t} = -\nabla\phi$ ∴ $E = -\nabla\phi - \frac{\partial A}{\partial t}$

scalar potential

(1) → $-\nabla^2\phi - \frac{\partial}{\partial t}\nabla \cdot A = \frac{1}{\varepsilon_0}\rho$

(6) → $\frac{1}{\mu_0}\nabla \times (\nabla \times A) + \varepsilon_0 \frac{\partial}{\partial t}\left( \nabla\phi + \frac{\partial A}{\partial t} \right) = J$

$-\frac{1}{\mu_0}\left( \nabla^2 A - \mu_0\varepsilon_0 \frac{\partial^2 A}{\partial t^2} \right) - \frac{1}{\mu_0}\nabla \left( \nabla \cdot A + \varepsilon_0\mu_0 \frac{\partial \phi}{\partial t} \right) = J$



COLLEGE NOTES / n075
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## Potential の任意性

変換　Scalar 場 $\Lambda(x,t)$

$$\left.\begin{aligned} A &\to A' = A + \nabla\Lambda \\ \phi &\to \phi' = \phi - \frac{\partial\Lambda}{\partial t} \end{aligned}\right\} \quad (7)$$

ゲージ変換

$$B' = \nabla\times A' = \nabla\times A = B$$

$$E' = -\nabla\phi' - \frac{\partial A'}{\partial t}$$
$$= -\nabla\phi - \frac{\partial A}{\partial t} + \underbrace{\nabla\frac{\partial\Lambda}{\partial t} - \frac{\partial}{\partial t}\nabla\Lambda}_{=0} = E$$

E, B は不変 ∴ この自由度で簡単化する (Lorentz 条件)

もし (8) をみたさなければ

$$\nabla\cdot A + \varepsilon_0\mu_0\frac{\partial\phi}{\partial t} = 0 \qquad (8)$$

$$(7) \to \left(\nabla^2\Lambda - \varepsilon_0\mu_0\frac{\partial^2\Lambda}{\partial t^2}\right) + \left(\nabla\cdot A + \varepsilon_0\mu_0\frac{\partial\phi}{\partial t}\right) = 0$$

スカラー　　となる $\Lambda$ を用いれば 必ず (8) が成り立つ

$$\begin{cases} \dfrac{1}{c^2}\dfrac{\partial^2\phi}{\partial t^2} - \nabla^2\phi = \dfrac{1}{\varepsilon_0}\rho & (9) \\ \dfrac{1}{c^2}\dfrac{\partial^2 A}{\partial t^2} - \nabla^2 A = \mu_0 J & (10) \\ \dfrac{1}{c^2}\dfrac{\partial\phi}{\partial t} + \nabla\cdot A = 0 \end{cases}$$

・ $\dfrac{1}{c^2}\dfrac{\partial^2\Lambda}{\partial t^2} - \nabla^2\Lambda = 0$

をみたす $\Lambda$ を作って

・ Coulomb gauge

$\nabla\cdot A = 0$

## 物質中

分極　$\rho_p = -\nabla\cdot P$

時間変化　$\dfrac{\partial\rho_p}{\partial t} \to$ current $J_p$

$= -\nabla\cdot J_p$

$$\left.\begin{aligned} J_p &= \frac{\partial P}{\partial t} \end{aligned}\right\}$$

$$\nabla\cdot E = \frac{1}{\varepsilon_0}(\rho_f - \nabla\cdot P)$$

$J = J_f + J_p + J_M$ より

$$\frac{1}{\mu_0}\nabla\times B = J_f + \frac{\partial P}{\partial t} + \nabla\times M + \varepsilon_0\frac{\partial E}{\partial t}$$

$$\to \nabla\times\underbrace{\left(\frac{1}{\mu_0}B - M\right)}_{H} = J_f + \frac{\partial}{\partial t}\underbrace{(\varepsilon_0 E + P)}_{D}$$

磁化　$M(x,t)$

$J_M = \nabla\times M$

$\nabla\cdot J_M = 0 \quad \therefore \rho_M = 0$

$D = \varepsilon_0 E + P$ , $H = \dfrac{1}{\mu_0}B - M$

$$\begin{cases} \nabla\cdot D = \rho \\ \nabla\times E = -\dfrac{\partial B}{\partial t} \\ \nabla\times H = J + \dfrac{\partial D}{\partial t} \\ \nabla\cdot B = 0 \end{cases}$$

## §3 電磁場のエネルギー，運動量　　保存則 P, E, L

$\varepsilon(x,t)$　　$\mathcal{E} = \int \varepsilon \, dV$

Charge　　Flux

$\rho \rightarrow J$

B·(Ⅲ)　$\frac{1}{\mu_0} B \cdot (\nabla \times E + \frac{\partial B}{\partial t}) = 0$

E·(Ⅱ)　$E \cdot (\frac{1}{\mu_0} \nabla \times B - \varepsilon_0 \frac{\partial E}{\partial t}) = J \cdot E$

$\Rightarrow \frac{1}{\mu_0} \nabla \cdot (E \times B) + \frac{1}{2} \frac{\partial}{\partial t} \left( \varepsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) = - J \cdot E$

$\varepsilon(x,t) = \frac{1}{2} \left( \varepsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$　… (1)

$P_\varepsilon(x,t) = \frac{1}{\mu_0} (E \times B)$　… (2)

c.f. 弦　$\varepsilon = \frac{1}{2} \sigma \left( \frac{\partial u}{\partial t} \right)^2 + \frac{1}{2} F \left( \frac{\partial u}{\partial x} \right)^2$

energy flux　$\frac{\partial \varepsilon}{\partial t} = - \frac{\partial}{\partial x} \left( F \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} \right)$

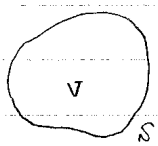
$\frac{\partial \varepsilon}{\partial t} + \nabla \cdot P_\varepsilon = - J \cdot E$　… (3)

$\varepsilon(x,t)$：エネルギー密度

$P_\varepsilon(x,t)$：Poynting ベクトル (1884)　Joule / sec. $m^2$

CGS
$\varepsilon = \frac{1}{8\pi} (E^2 + B^2)$
$P_\varepsilon = \frac{c}{4\pi} E \times B$

$$\mathcal{E} = \int_V \varepsilon(x,t) \, dx$$




(3)　$\frac{d\mathcal{E}}{dt} = - \int_S P_s \cdot dS - \underbrace{\int_V J \cdot E \, dx}$　場が粒子にする仕事　☺ Lorentz 力 $f_i = g_i E + g_i v_i \times B$

(Joule 熱)

運動量の保存　($P_m = \sum P_i$ 粒子の運動量)

$\frac{dP_m}{dt} = \sum_i f_i = \int_V \{ \rho(x,t) E(x,t) + J \times B \} dx$

$= \int_V \{ \varepsilon_0 (\nabla \cdot E) E + (\frac{1}{\mu_0} \nabla \times B - \varepsilon_0 \frac{\partial E}{\partial t}) \times B \} dx$

$= \underbrace{- \int_V \frac{\partial}{\partial t} (\varepsilon_0 E \times B) dx} + \int_V \{ \varepsilon_0 [(\nabla \cdot E) \cdot E - E \times (\nabla \times E)] + \frac{1}{\mu_0} (\nabla \times B) \times B \} dx$

$- \frac{d}{dt} \underbrace{\int_V \varepsilon_0 \mu_0 \vec{P}_\varepsilon \, dx}$

$\vec{P}_f$：場の運動量

$\frac{d}{dt} (P_m + P_f)$ ↙

$\frac{\partial E}{\partial t} \times B = \frac{\partial (E \times B)}{\partial t} - E \times \frac{\partial B}{\partial t}$

$= \quad " \quad + E \times (\nabla \times E$

$\{\}_i = \varepsilon_0 [\partial_j E_j E_i - \epsilon_{ijk} E_j \epsilon_{klm} \partial_l E_m] + \frac{1}{\mu_0} \epsilon_{ijk} \epsilon_{jlm} \partial_l B_m B_k$

$( \epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl} )$

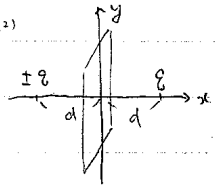
$\{\}_i = \partial_j T_{ij}(x,t)$ ：$T_{ij} = \varepsilon_0 E_i E_j + \frac{1}{\mu_0} B_i B_j - \frac{1}{2}\left(\varepsilon_0 E_k E_k + \frac{1}{\mu_0} B_k B_k\right)\delta_{ij}$

Maxwell の応力テンソル　　2階の対称テンソル

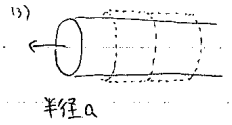
$$\frac{d}{dt}(P_m + P_f)_i = \int_V \partial_j T_{ij}\, dx = \int_V T_{ij}\, dS_j$$

↑ Vの内の Total moment

例題 (2)




y-z 面に働く力を求めよ。　(1) $T_{ij}$ の成分を求めよ




定常電流 $j = \sigma E$

(3) を適用

$P_E = \frac{1}{\mu_0} E \times B$　　photon $c = \frac{\omega}{k}$

運動量 $\varepsilon_0 \mu_0 P_E = \frac{1}{c^2} P_E$　　energy flow $\varepsilon \cdot c = \hbar\omega \cdot c = \hbar k \cdot c^2 = pc^2$

- 媒質中の波
  光学
- 電磁波の放出
- 特殊相対論

# 第8章 真空中の電磁波

## §1 波動方程式

$\rho = 0$, $J = 0$　　static な場はありえない

$$\left.\begin{aligned} \frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2} - \nabla^2\phi &= 0 \\ \frac{1}{c^2}\frac{\partial^2 A}{\partial t^2} - \nabla^2 A &= 0 \end{aligned}\right\} \cdots (1)$$

$\Box\phi = 0$, $\Box A = 0$ ) wave eq.

Lorentz gauge $\frac{1}{c^2}\frac{\partial\phi}{\partial t} + \nabla A = 0$

$$\left(\begin{aligned} E &= -\nabla\phi - \frac{\partial A}{\partial t} \\ B &= \nabla \times A \\ \varepsilon_0\mu_0 &= \frac{1}{c^2} \end{aligned}\right) \cdots (2)$$

制限された gauge Transt

$\Lambda$ も (1) をみたす

$A \to A' = A + \nabla\Lambda$　　$(\Box\Lambda = 0)$

$\phi \to \phi' = \phi - \frac{\partial\Lambda}{\partial t}$

$$\Lambda(x,t) = \int^t \phi(x,t')\,dt'$$

$\phi = 0$, $\nabla A = 0$ ⋯(3)

平面波　d'Alembert の解

$$\left(\frac{1}{c}\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right)\left(\frac{1}{c}\frac{\partial}{\partial t} + \frac{\partial}{\partial x}\right) A_i(x,t) = 0 \qquad (4)$$

$\xi = x - ct$, $\eta = x + ct$

(4) $\to$ $\frac{\partial^2 A_i}{\partial\xi\partial\eta} = 0$ $\Rightarrow$ $A_i = f_i(\xi) + f_2(\eta)$ ⋯(5)

右に進む波のみ

$A = A(x - ct)$

(3) $\to$ $\frac{\partial A_x}{\partial x} = 0$　　$A_x = 0$ ととることができる.　$\hat{x}\cdot A = 0$

$E = -\frac{\partial A}{\partial t}$ $\to$ $E = cA'$

$'$ は $\xi$ についての微分

$\hat{x}\cdot E = 0$, $B = \nabla\times A = \frac{1}{c}\hat{x}\times E$

進行方向 $\hat{n}$

$E = E(\hat{n}x - ct)$　　$\hat{n}\cdot E = 0$

$B = B(\;\;''\;\;)$　　$\hat{n}\cdot B = 0$　　$B = \frac{1}{c}\hat{n}\times E$　　$E\times B = \hat{n}E^2/c = c\hat{n}B^2$

横波 transverse wave

Poynting vecter

$$P_E = \frac{1}{\mu_0} E\times B = \frac{1}{\mu_0}\hat{n}\frac{1}{2}\left(\frac{E^2}{c} + cB^2\right) = \frac{c}{2}\hat{n}\left(\varepsilon_0 E^2 + \frac{1}{\mu_0}B^2\right) \cdots (8)$$

速度 c でエネルギーの流れ

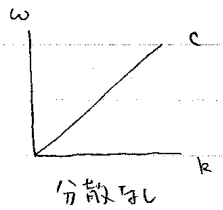
## §2 単色平面波 monochromatic ωの平面波

複素表示

$A = \mathrm{Re}\, A_0 e^{i(k\cdot x - \omega t)}$ $k$: 波数ベクトル (1)

$|A_0|$ 振幅 $k = \frac{\omega}{c}\hat{n}$ (2)

波長 $\lambda = 2\pi/k$



分散なし

$E = i\omega A$
$B = ik \times A$ (3)

$B = \frac{1}{\omega} k \times E$

一般の単色平面波の形

$E = \mathrm{Re}\, E_0 e^{i(k\cdot x - \omega t)}$ $E_0 = E_{01} + iE_{02}$ $E = E_{01}\cos(k\cdot x - \omega t) - E_{02}\sin(k\cdot x - \omega t)$ … (4)

$k\cdot E = 0$ , $k\cdot E_{01} = k\cdot E_{02} = 0$ … (5)

もし $E_{01}\cdot E_{02} \neq 0$

$\tan 2\alpha = \frac{2E_{01}E_{02}}{E_{01}^2 - E_{02}^2}$ できる角度 $\alpha$ を用いて

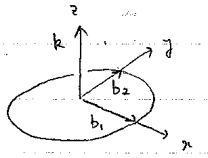
$b_1 = E_{01}\cos\alpha + E_{02}\sin\alpha = \mathrm{Re}(E_0 e^{-i\alpha})$
$b_2 = -E_{01}\sin\alpha + E_{02}\cos\alpha = \mathrm{Im}(E_0 e^{-i\alpha})$ … (6)

$E = b_1\cos(k\cdot x - \omega t + \alpha) + b_2\sin(k\cdot x - \omega t + \alpha)$ … (7)

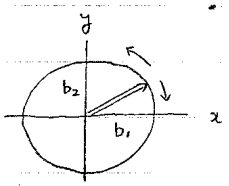
$b_1\cdot b_2 = 0$ , $b_1 \times b_2 = E_{01} \times E_{02}$

$E_{01}\times E_{02}$ の方向 $\pm k$

⊕ 右まき
⊖ 左まき
helicity $\pm 1$




$\frac{E_x^2}{b_1^2} + \frac{E_y^2}{b_2^2} = 1$



楕円偏光

$b_1 = b_2$ 円偏光

$b_2 = 0$ 直線偏光 $E_0(\hat{x} + i\hat{y})e^{i\phi}$

波の平均強度 $\overline{E^2} = \frac{1}{2}(E_{01}^2 + E_{02}^2)$

$= \frac{1}{2}(b_1^2 + b_2^2)$

電磁波にともなう E

sunlight $1\,\mathrm{kW/m^2}$

$\left(\begin{array}{l} 4\pi\epsilon_0 = 10^7 c^{-2} \\ \mu_0 = 4\pi\, 10^{-7} \end{array}\right)$ $P_\varepsilon = \frac{c}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right)$

円偏光 $E = E_0(\hat{x}\cos\omega t + i\hat{y}\sin\omega t)$ → $\overline{E^2} = E_0^2$

$\bar{P}_\varepsilon = c\epsilon_0 E_0^2 = \frac{10^7}{4\pi c} E_0^2$

$W/m^2 = \mathrm{Joule/s\cdot m^2}$ $E_0 = 19.4\sqrt{\bar{P}_\varepsilon}$ $\bar{P}_\varepsilon = 1\,W/m^2$ , $E_0 \sim 19.4$ Volt/m

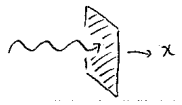
Laser $100\,W/mm^2$

一般に $\overline{\underset{\text{real}}{A(t)B(t)}} = \overline{A(t)B^*(t)}$

No. 31

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輻射圧 Radiation Pressure




圧力 $T_{xx}$ 成分

$P_F = \frac{1}{c^2} P_E$

圧力 $cP_F = \frac{1}{c} P_E$

Sunlight $3 \cdot 10^{-6}\ N/m^2$

スペクトル分解

離散的

基本周期 $T$ $\quad \omega_0 = \frac{2\pi}{T}$

$$f(t) = \sum_{n=-\infty}^{\infty} f_n e^{-in\omega_0 t}$$

$$f_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t) e^{in\omega_0 t}\, dt$$

$f$ real $\rightarrow f_n^* = -f_n$

$$\overline{f^2(t)} = \sum_{n=-\infty}^{\infty} f_n^* f_n$$

フーリエ積分

$$f(t) = \int_{-\infty}^{\infty} f_\omega e^{-i\omega t} \frac{d\omega}{2\pi}$$

$$f_\omega = \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt$$

複素表示

$\overline{E^2}_{real} = \frac{1}{2} E_0^* E_0$

$\overline{E^* E} = \quad "$

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Date 93. 11. 5

# 第9章

## §1 媒質中の電磁波

等方的　　マクロの取扱い

D, H　　$\varepsilon$, $\sigma$　複素 $\sigma(\vec{k},\omega)$

$\nabla D = \rho_f$ ,　$D = \varepsilon E$　　$(D = \varepsilon_0 E + P)$　　$kc = \omega$

$\nabla \times E = -\frac{\partial B}{\partial t}$ ,　$B = \mu H$　　$(H = \frac{1}{\mu_0} B - M)$　　線形 → 重ね合せ可能 → フーリエ分解

$\nabla \times H = J + \frac{\partial D}{\partial t}$　　$J = \sigma_e E$

$\nabla \cdot H = 0$

分極モデル（振動子）

$m\ddot{z} = -m\omega_0^2 z - m\gamma\dot{z} + eE_M$　　$E_M \propto e^{-i\omega t}$

$z = Z_\omega e^{-i\omega t}$

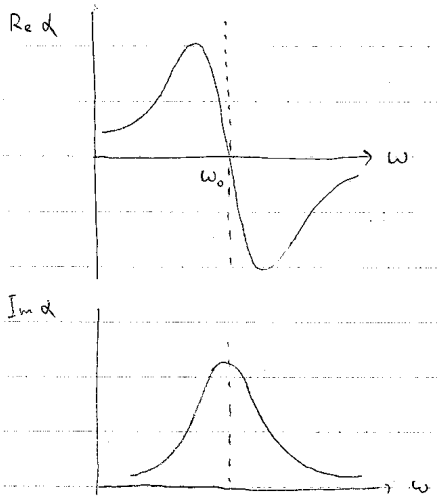
個々の分子

$nP = P = n\alpha E_M$　　$\alpha$：分極率

$p = ez$

$$\varepsilon = \frac{1 + 2n\alpha/3\varepsilon_0}{1 - n\alpha/3\varepsilon_0}\,\varepsilon_0$$

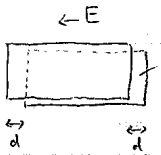
$$\alpha(\omega) = -\frac{e^2}{m}\cdot\frac{1}{\omega^2 - \omega_0^2 + i\gamma}$$




$\mu \sim \mu_0$

導体　自由電子

$\rho_f \simeq 0$




表面電荷

$\sigma = \pm ned$

自由電子の密度

この分極に $E \sim \frac{1}{\varepsilon_0} ned$

全ての電荷に働く

電子

$-N \cdot \frac{1}{\varepsilon_0} ned \cdot e = mN\ddot{d}$

$$\boxed{\omega_{pl}^2 = \frac{ne^2}{\varepsilon_0 m}}$$

$\left( n \sim 10^{23} \quad e \sim 1.6 \cdot 10^{-19} \right.$

$\left. \varepsilon_0 \sim 9 \cdot 10^{-12} \quad m \sim 9 \cdot 10^{-31} \right)$

$\omega_{pl} \sim 10^{16}\ \mathrm{sec}^{-1}$

Atm　$\omega_0 \sim e\nabla$

$10^{16}\ \mathrm{sec}^{-1}$

$Na^+Cl^-$　$10^{14}\ \mathrm{sec}^{-1}$

$\rho_f \neq 0$ decay の時間

$\frac{\partial \rho_f}{\partial t} + \nabla \cdot J = 0$　　$\nabla \cdot J = \sigma_e \nabla \cdot E = \frac{\sigma_e}{\varepsilon_0} \nabla \cdot D = \frac{\sigma_e}{\varepsilon_0} \rho_f$

$\frac{\partial \rho_f}{\partial t} + \frac{\sigma_e}{\varepsilon_0} \rho_f = 0$　　$\rho_f \propto e^{-t/\tau}$

$\tau = \frac{\varepsilon_0}{\sigma_e}$

Cu　$\sigma_e \sim 6 \times 10^7\ (\Omega m)^{-1}$ ; $\tau \sim 10^{-19}$ (sec)

海水　$\sim 5$　　; $\sim 10^{-12}$

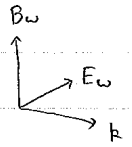
# 単色平面波

$E = E_\omega e^{i(kx-\omega t)}$

$B = B_\omega$ 〃

$k \cdot E_\omega = 0$ 　　$k, E_\omega, B_\omega$ は互いに直交

$k \times E_\omega = \omega B_\omega$




$-\frac{1}{\mu} k \times B_\omega = i\sigma_e E_\omega + \omega \varepsilon E_\omega \rightarrow -k^2 B_\omega = -\omega(i\sigma\mu + \omega\varepsilon\mu) B_\omega$

$\omega$ : real
$k$ : complex

屈折率 $n = kc/\omega$ を導入

$$n = \left[ \frac{\varepsilon\mu + i\mu\sigma_e/\omega}{\varepsilon_0 \mu_0} \right]^{1/2}$$

例 (1) 誘電体 　$\mu = \mu_0$ 　$\sigma_e$ small

$n \sim \sqrt{\varepsilon/\varepsilon_0} \sim 1$

(2) 導体 　$\varepsilon \sim \varepsilon_0 \sim 10^{-11}$

$$n \sim \left[ \frac{i\mu\sigma_e}{\varepsilon_0 \mu_0 \omega} \right]^{1/2} \qquad i^{\frac{1}{2}} = \frac{1}{\sqrt{2}}(1+i)$$

$n = n' + in''$

$E(x,t) = E_\omega e^{in'\frac{\omega}{c}k\cdot x} e^{-n''\frac{\omega}{c}k\cdot x} \rightarrow \delta = \frac{c}{n''\omega}$ : Skin depth

$$= \frac{c}{\frac{1}{\sqrt{2}}\sqrt{\frac{\mu\sigma_e}{\varepsilon_0\mu_0\omega}}\cdot\omega} = \sqrt{\frac{2}{\mu\sigma_e\omega}}$$

$\omega \sim 2\times10^{10}$ 　Cu 　$\delta \sim 10^{-6}$ cm

海水 $\sim 75$ Hz 　$\delta \sim 28$ m 　せんすいかんでは長波を使う

## §2 光学活性 Optical Activity

No.J 12

$E = E_{01} \cos(k\cdot x - \omega t + \alpha) - E_{02} \sin(k\cdot x - \omega t + \alpha)$

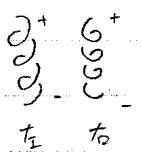
$k \perp E_{01} \perp E_{02}$

$E_{01} \times E_{02} \parallel k$

等方的

分子

結晶固体 $SiO_2$




Handedness




$P$ にともなう $J \rightarrow M$

半径 $a$ 　$e\xi(t) \rightarrow P$

ヒゲ 4本 　うせんにそって動く速度

$v = 2\pi a / \left(\frac{b}{v_z}\right)$

$m = \pm I \pi a^2 N$ 巻数

$= \pm \frac{\pi a^2}{b} N \frac{dP}{dt}$

$$M(x,t) = \eta \cdot \frac{d\mathbb{P}(x,t)}{dt} \qquad (1)$$

$\eta$ : 長さの次元の物理定数

$$H = \frac{B}{\mu_0} - M$$

$$\nabla \times H = \nabla \times \left( \frac{B}{\mu_0} - \eta \frac{\partial P}{\partial t} \right) = \frac{\partial \varepsilon E}{\partial t} \qquad (2)$$

$$P = (\varepsilon - \varepsilon_0) E$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\frac{1}{\mu_0} \nabla^2 E^2 - \varepsilon \frac{\partial^2 E}{\partial t^2} = \eta (\varepsilon - \varepsilon_0) \nabla \times \frac{\partial^2 E}{\partial t^2} \qquad (3)$$

単色平面波

$$\frac{1}{\mu_0} (-k^2 + \mu_0 \varepsilon \omega^2) E_\omega = -i \eta (\varepsilon - \varepsilon_0) \omega^2 k \times E_\omega$$

$$\rightarrow \frac{1}{\mu_0^2} (k^2 - \mu_0 \varepsilon \omega^2)^2 = (\eta (\varepsilon - \varepsilon_0) \omega^2 k)^2 \qquad \therefore k^2 - \mu_0 \varepsilon \omega^2 = \pm \eta \mu_0 (\varepsilon - \varepsilon_0) \omega^2 k \qquad (4)$$

(3) → $E = \pm i k \times E$ (5) $\quad \hat{k} = \hat{z}$, $E = E_x \hat{x} + E_y \hat{y}$ $\quad E_x = \pm i E_y$

(4) で $\eta k \ll 1$ のとき $\quad n_0 = kc/\omega$

$$\boxed{n_L - n_R \cong \eta \left( \frac{\varepsilon}{\varepsilon_0} - 1 \right) \frac{\omega}{c}} \qquad (6)$$

$$E_L = \frac{1}{2} E_\omega (\hat{x} + i\hat{y}) e^{i(n_L z/c - t)\omega}$$
$$E_R = \frac{1}{2} E_\omega (\hat{x} - i\hat{y}) e^{i(n_R z/c - t)\omega} \qquad (7)$$

$z = 0$ で直線偏光 $\parallel \hat{x}$ $\quad E = E_L + E_R$ $\quad$ 偏光面の回転

単位長さ あたりの $\theta$ Specific Rotatory Power

$$(7) = \frac{\pi}{\lambda} (n_L - n_R) \qquad (8)$$

Green light

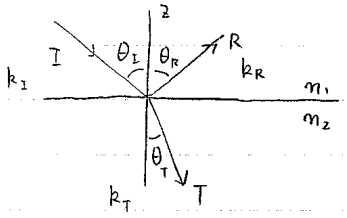
| | |
|---|---|
| Quartz | 220°/cm |
| Syrup | 10°/cm |
| 液晶 | 大 |

Faraday 効果 $\quad \uparrow B$ $\quad \uparrow \hat{k}$

$$n_R - n_L = \frac{\lambda}{\pi} V B$$

↑ $H_2O$ $\quad 1.31 \times 10^{-2}$ min /Gauss

## §3 反射 (Reflection) と屈折 (Refraction)




境界条件

a) $(\mathbb{D}_2 - \mathbb{D}_1)\,\hat{n} = \sigma_f$
b) $(\mathbb{B}_2 - \mathbb{B}_1)\,\hat{n} = 0$
c) $(\mathbb{E}_2 - \mathbb{E}_1)\,\hat{t} = 0$
d) $(\mathbb{H}_2 - \mathbb{H}_1)\,\hat{t} = j(\hat{n} \times \hat{t})$ } (1)

$\mathbb{E}_\alpha = \mathcal{E}_\alpha e^{i(k_\alpha \cdot x - \omega_\alpha t)}$
($\alpha = I, R, T$ waves)
$\mathbb{B}_\alpha = \frac{n_\alpha}{c} k_\alpha \times \mathbb{E}_\alpha$
($n_I = n_R = n_1$, $n_T = n_2$) } (2)

Time $\omega_\alpha = \omega$ (const)

Space $k_I x = k_R \cdot x = k_T \cdot x$
($x$ : 境界面中のベクトル)

$k_I = k_R$, $\frac{k_I}{n_1} = \frac{k_T}{n_2}$ (3) ($n = kc/\omega$)

$k_I \hat{k}_I \cdot x = k_I \hat{k}_R x = \frac{n_2}{n_1} k_I \hat{k}_T x$ $\hat{k}_I x = \hat{k}_R x$ $k_I \hat{n} = -k_R \hat{n}$

$\therefore \theta_R = \theta_I$

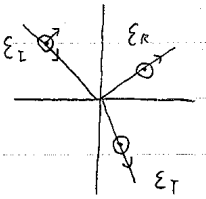
$n_1 \sin\theta_I = n_2 \sin\theta_T$ (Snell の法則) (4)

散乱面 = $k_I$ と $\hat{n}$ の作る面

3つのベクトル $k_I, k_R, k_T$ はすべてこの面内

a) $\varepsilon_1 (\mathcal{E}_I + \mathcal{E}_R)\hat{n} = \varepsilon_2 \mathcal{E}_T \hat{n}$
b) 〃
c) $(\mathcal{E}_I + \mathcal{E}_R) \cdot \hat{t} = \mathcal{E}_T \hat{t}$
d) $(\mathbb{B}_I + \mathbb{B}_R) \cdot \hat{t} = \mathbb{B}_t \hat{t}$

偏光の方向 = 電場ベクトルの方向




① 散乱面 ⊥
② 〃

① $\left(\frac{\mathcal{E}_T}{\mathcal{E}_I}\right)_\perp = \frac{2n_1 \cos\theta_I}{n_1 \cos\theta_I + n_2 \cos\theta_T}$ Fresnel の式

$\left(\frac{\mathcal{E}_R}{\mathcal{E}_I}\right)_\perp = \frac{n_1 \cos\theta_I - n_2 \cos\theta_T}{n_1 \cos\theta_I + n_2 \cos\theta_T}$ 〃

② $\left(\frac{\mathcal{E}_T}{\mathcal{E}_I}\right)_{//} = \frac{2n_1\cos\theta_I}{n_1\cos\theta_T + n_2\cos\theta_I}$

$\left(\frac{\mathcal{E}_R}{\mathcal{E}_I}\right)_{//} = \frac{n_1\cos\theta_T - n_2\cos\theta_I}{}$

(例). 垂直入射 $\theta_I = 0$ $\quad \frac{\mathcal{E}_T}{\mathcal{E}_I} = \frac{2n_1}{n_1+n_2}$ $\quad \frac{\mathcal{E}_R}{\mathcal{E}_I} = \frac{n_1-n_2}{n_1+n_2}$

・全反射 $\theta_I = \frac{\pi}{2}$ $\quad \mathcal{E}_R = -\mathcal{E}_I$

反射率 $R = \left(\frac{\mathcal{E}_R}{\mathcal{E}_I}\right)^2$ $\quad R = \frac{P_n^{(R)}}{P_n^{(I)}}$

透過率 $T = \frac{n_2\cos\theta_T}{n_1\cos\theta_I}\left(\frac{\mathcal{E}_T}{\mathcal{E}_I}\right)^2$ $\quad R + T = 1$

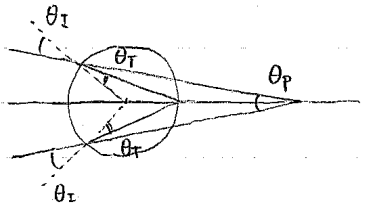
偏光面

$$R_{//} = \left(\frac{n_1\cos\theta_T - n_2\cos\theta_I}{n_1\cos\theta_T + n_2\cos\theta_I}\right)^2$$

§4 虹I

$\pi - 2\theta_T + \theta_I + \frac{\theta_P}{2} = \pi$

$\frac{1}{2}\theta_P = 2\theta_T - \theta_I$ (1)




Snell

$\sin\theta_I = n\sin\theta_T$ (2)

(1),(2)

$\frac{d\theta_P}{d\theta_I} = 4\frac{d\theta_T}{d\theta_I} - 1 = \frac{4}{n}\cdot\frac{\cos\theta_I}{\cos\theta_T} - 2$ (3)

$\frac{d\theta_P}{d\theta_I} = 0$ のとき intensity 最大

$\sin\theta_I = \frac{1}{n}\sqrt{\frac{4-n^2}{3}}$

violet ~ red

1.344 1.331

水 $n \sim \frac{4}{3}$ → $\theta_{P\,Max} = 42°$

$\theta_T = 40.2°$

cf. $\theta_B = 36.9°$ ブリュースター

2次の虹I

$\theta_S = \pi + 2(\theta_I - 3\theta_T)$

$\sim 51.1°$

## §5 導波管 wave guide



skin depth $\delta \sim \sqrt{\frac{2}{\sigma_c \mu \omega}}$

誘電体 $\sigma_c$ 小、

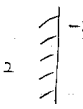
導体 $n = \left[\frac{\varepsilon\mu + i\mu\sigma_c/\omega}{\varepsilon_0 \mu}\right]^{1/2}$ $\omega = ck$

$|m| \sim \frac{1}{k\delta}$

Ag $\delta \sim 5\times10^{-6}$ cm

$\omega \sim 2\times10^{10}\,\mathrm{sec}^{-1}$

metal の内部 $E=0$ , $B=0$



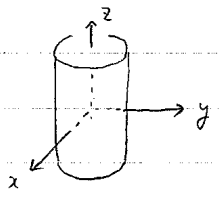

境界条件 a)~d)

a) $E_n = \sigma_f / \varepsilon_0$

b) $B_n = 0$

c) $\mathbf{E}\cdot\hat{t} = 0$

d) $\mathbf{B}\cdot\hat{t} = -\mu_0 j_f (n\times t)$




z方向に propagats

$$\begin{cases} \mathbf{E} = \mathcal{E}(x,y)e^{i(kz-\omega t)} \\ \mathbf{B} = \mathcal{B}(x,y) \quad \text{〃} \end{cases} \qquad (1)$$

空洞の中

$\sigma=0$, $J=0$ $\quad \nabla\times(\nabla\times E) = -\nabla\times\frac{\partial B}{\partial t} \rightarrow -\nabla^2 \mathbf{E} = -\frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}$ (2)

Bも同様

(1)に代入

$$(k^2 - \Delta_2)\mathcal{E}_z = \frac{\omega^2}{c^2}\mathcal{E}_z \rightarrow \begin{cases} \left[\Delta_2 + \frac{\omega^2}{c^2} - k^2\right]\mathcal{E}_z = 0 \\ [\quad\text{〃}\quad]\mathcal{B}_z = 0 \end{cases} \qquad (3)$$

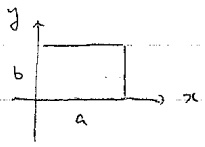
もとえば

$\nabla\times\mathbf{B} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$ $\qquad$ $\nabla\times\mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

$-\frac{i\omega}{c^2}\mathcal{E}_x = \partial_y \mathcal{B}_z - ik\mathcal{B}_y$ $\qquad$ $ik\mathcal{E}_x - \partial_x \mathcal{E}_z = i\omega \mathcal{B}_y$

$$\left(\frac{\omega^2}{c^2} - k^2\right)\mathcal{E}_x = i\omega\,\partial_y \mathcal{B}_z + ik\,\partial_x \mathcal{E}_z$$

$$(\quad\text{〃}\quad)\mathcal{E}_y = -i\omega\,\partial_x \mathcal{B}_z + ik\,\partial_y \mathcal{E}_z$$




どの側面でも $\mathcal{E}_z = 0$

$\mathcal{E}_x = 0$ , $y = 0, b$

$\mathcal{E}_y = 0$ , $x = 0, a$

但し $\frac{\omega^2}{c^2} - k^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

$\omega_{m,n}$

$\mathcal{E}_z = \mathcal{E} \sin \frac{\pi m}{a} x \sin \frac{\pi n}{b} y$

$B_z = B \cos \frac{\pi m}{a} x \cos \frac{\pi n}{b} y$

a) TM mode (Transverse Magnetic)

$B_z = 0$ $\mathcal{E}_z \neq 0$ $m, n$ ともに 0 ではない

$m = 1, n = 1$

$\frac{\omega_{11}^2}{c^2} = k^2 + \left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2$

$k^2 = \frac{\omega_{11}^2}{c^2} - [\ 〃\ ]$

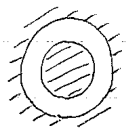
b) TE mode

$\mathcal{E}_z = 0$, $B_z \neq 0$

$(1,0), (0,1)$

$TE_{10}$ $\mathcal{E}_x = B_y = 0$

$\frac{\omega}{k} = c\sqrt{1 + \frac{\pi^2}{k^2}\left(\frac{1}{a^2} + \frac{1}{b^2}\right)}$

問  同軸ケーブル

位相速度 phase velocity

# 第10章 電磁波の放出

波動 完全流体

密度波

$$v(x,t) = -\nabla\phi(x,t)$$

$$\frac{\partial^2\phi}{\partial t^2} - s^2\nabla^2\phi = 0$$

球面波 $\phi = \phi(r,t)$

$$\phi = \frac{f(t\mp r/s)}{r} = \frac{1}{4\pi\rho_0}\cdot\frac{Q(t-r/s)}{r}$$

source $Q(t)$

$Q = Q_\omega e^{-i\omega t}$

$$v_r = -\frac{\partial\phi}{\partial r} = \frac{f}{r^2} + \frac{f'}{sr}$$

$$v_r = -\frac{Q_\omega k}{4\pi\rho_0}\cdot\frac{\sin(\omega t-kr)}{r} + \frac{Q_\omega}{4\pi\rho_0}\frac{1}{r^2}\cos(\omega t-kr)$$

$kr \gg 1$ 波動帯

## §1 遅延ポテンシャル (Retarded Potential)

$$\begin{cases} E = -\nabla\phi - \dfrac{\partial A}{\partial t} \\ B = \nabla\times A \end{cases} \quad (1)$$

Lorentz gauge

$$\nabla\cdot A + \varepsilon_0\mu_0\frac{\partial\phi}{\partial t} = 0 \quad (2)$$

Maxwell より

$$\begin{cases} -\dfrac{1}{c^2}\dfrac{\partial^2\phi}{\partial t^2} + \nabla^2\phi = -\dfrac{1}{\varepsilon_0}\rho \\ -\dfrac{1}{c^2}\dfrac{\partial^2 A}{\partial t^2} + \nabla^2 A = -\mu_0 J \end{cases} \quad (3)$$

問題 $\rho, J$ given $\longrightarrow$ $A, \phi$

すたようない とき

$$\begin{cases} \phi(x) = \dfrac{1}{4\pi\varepsilon_0}\displaystyle\int\frac{\rho(x')}{|x-x'|}dx' \\ A(x) = \dfrac{\mu_0}{4\pi}\displaystyle\int\frac{J(x')}{|x-x'|}dx' \end{cases}$$

(3) すべて wave eqn

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)F(x,t) = -S(x,t) \quad (4)$$

$S(x,t)$ given $\longrightarrow$ $F$ ?

1) 点電荷の分布

重ね合わせ

$$S(x,t) = \delta(x)\cdot S(t)$$

このとき $F(r,t)$

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r}\right)F(r,t) = -\delta(x)\cdot S(t)$$

$x \neq 0$ のとき $= 3$

$$F = \frac{f(r,t)}{r}$$

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial r^2}\right)f = 0$$

d'Alembert の解　　Advanced potential

$$f(r,t) = f_R\left(t-\frac{r}{c}\right) + f_A\left(t+\frac{r}{c}\right)$$




原点のまわりで Gauss の定理

$$\int_V \nabla^2 F \, dx = \int_S \nabla F \cdot d\mathbf{S} = \int_S \frac{\partial}{\partial r}\left(\frac{f}{r}\right) r^2 d\Omega = 4\pi \left(r\frac{\partial f}{\partial r} - f\right)_{r=R}$$

$$\lim_{R\to 0} -4\pi f(R,t) = -\int_V s(t)\,\delta(x)\,dx = -s(t) \qquad f(0,t) = \frac{1}{4\pi} s(t)$$

$$f_R(t) + f_A(t) = \frac{1}{4\pi} s(t)$$

$$F(x,t) = \frac{1}{4\pi r}\left\{ C_R\, s\left(t-\frac{r}{c}\right) + C_A\, s\left(t+\frac{r}{c}\right)\right\} \qquad (\text{但し } C_R + C_A = 1)$$

一般には

$$s(x,t) = \int s(x',t)\,\delta(x'-x)\,dx' \qquad \leftarrow x' に s(x',t) なる source$$

$$\rightarrow \boxed{F(x,t) = \frac{1}{4\pi}\int dx' \frac{s(x',\, t-R/c)}{R}} \qquad R = |x-x'| \qquad (5)$$

Causality から Retarded だけ

2) Fourier 変換の方法

$$F(k,\omega) = \int_{-\infty}^{\infty} dx\, dt\, F(x,t)\, e^{-i(k\cdot x - \omega t)}$$

δ-関数

$$\delta(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dp\, e^{ipx}$$

$$F(x,t) = \frac{1}{(2\pi)^4}\int_{-\infty}^{\infty} dk\, d\omega\, F(k,\omega)\, e^{i(k\cdot x - \omega t)}$$

$$s(x,t) = \int dx'dt' s(x',t') \int \frac{dk\,d\omega}{(2\pi)^4} e^{i(k(x-x') - \omega(t-t'))}$$

Wave eqn

$$\left(\frac{1}{c^2}\omega^2 - k^2\right) F(k,\omega) = -\int dx'\, dt'\, s(x',t')\, e^{-i(k\cdot x' - \omega t')}$$

$$F(x,t) = -\int \frac{dk\,d\omega}{(2\pi)^4} \int dx'dt\; S(x',t') \cdot \frac{e^{i\left[k(x-x') - \omega(t-t')\right]}}{\omega^2/c^2 - k^2} \qquad (6)$$

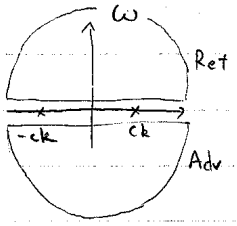
$$\int dk = \int_0^\infty k^2 dk \int_{-1}^{1} d\mu \int d\varphi$$

$|x-x'| \cdot k$

$$\rightarrow \frac{4\pi}{R} \int_0^\infty dk\, k \cdot \frac{\sin kR}{\omega^2/c^2 - k^2} e^{-i\omega(t-t')}$$

$$c^2 \int_{-\infty}^{\infty} d\omega \frac{1}{\omega^2 - c^2k^2} e^{-i\omega(t-t')}$$

$$\frac{1}{2ck}\left(\frac{1}{\omega - ck} + \frac{1}{\omega + ck}\right)$$




$\omega \pm i\epsilon$ + Retarded − Advanced

plus Homogenious solution

$t-t' > 0$ のとき 上の contour $\quad -2\pi i \frac{1}{2ck}\theta(t-t')\left[e^{-ick(t-t')} - e^{ick(t-t')}\right]$

$t-t' < 0$ 〃 〃 $\quad 2\pi i \frac{1}{2ck}\theta(t'-t)\left[\quad 〃 \quad\right]$

$$F_A(x,t) = F_{out}(x,t) + \int dx \frac{S(x', t + \frac{1}{c}R)}{|x-x'|}$$

$\neq 0$

$$F_R(x,t) = F_{in}(x,t) + \int dx \frac{S(x', t - \frac{1}{c}R)}{|x-x'|}$$

$= 0$

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Retarded

$$\frac{4\pi}{R}\int_0^\infty k\,dk\,\sin kR\;\; c\pi i^2\,\frac{1}{k}\,\theta(t-t')\,2\sin kc(t-t')$$

$$dk \to \delta(R-c(t-t'))$$

$$F_R(x,t) = \frac{1}{4\pi}\int dx'\,\frac{1}{R}\,S\left(x',\,t-\frac{R}{c}\right) \qquad R=|x-x'|$$

$$\begin{cases} \phi(x,t) = \dfrac{1}{4\pi\varepsilon_0}\displaystyle\int dx'\,\frac{1}{R}\,\rho\left(x',\,t-\frac{R}{c}\right) \\ A(x,t) = \dfrac{\mu_0}{4\pi}\displaystyle\int dx'\,\frac{1}{R}\,\mathbb{J}\left(x',\,t-\frac{R}{c}\right) \end{cases} \qquad (7)$$

Lorentz 条件

$$\frac{1}{c^2}\frac{\partial\phi}{\partial t} + \nabla\cdot A = \int dx'\left\{\frac{1}{R}\frac{\partial}{\partial t}\rho\left(x,t-\frac{R}{c}\right) + \nabla\cdot\left[\frac{\mathbb{J}(x,t-R/c)}{R}\right]\right\}$$

$R = x - x'$

$$-\nabla'\cdot[\;\;] + \frac{1}{R}\nabla''\cdot\mathbb{J}(x'',\,t-R/c)\Big|_{x''\to x'}$$

$[\;\;]$ ↓ 0

連続の方程式

O.K.

§3 単色波の放射 Dec 3

$$\left.\begin{aligned} \rho(x,t) &= \int \frac{\omega}{2\pi}\,\rho(x,\omega)\,e^{-i\omega t} \\ \mathbb{J}(x,t) &= \int \text{〃}\,\mathbb{J}(x,\omega)\;\text{〃} \end{aligned}\right\} \qquad (1)$$

$$A(x,\omega)\,e^{-i\omega t} = \int dx'\,\frac{\mathbb{J}(x',\omega)}{R}\,e^{-i\omega\left(t-\frac{R}{c}\right)}$$

$\omega = kc$




$$A(x,\omega) = \int dx'\,\frac{\mathbb{J}(x',\omega)}{R}\,e^{ikR} \qquad (2)$$

Pで $\rho$, $\mathbb{J}$ が 0

$$\left.\begin{aligned} B &= \nabla\times A \\ \frac{\partial E}{\partial t} &= c^2\nabla\times B \end{aligned}\right\} \qquad (3)$$

Aだけから場は決定される

Scale

① $\omega$ $\quad \lambda = \frac{2\pi}{k} = \frac{2\pi}{\omega}$

② source の size $d$

原子・分子

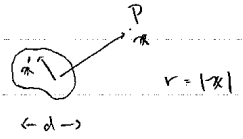
$\lambda \sim 10^{-4}$ cm

$d \sim 10^{-8}$ cm

$\lambda \gg d$

アンテナ

kMHz $\quad \lambda \sim 30$ cm




$d \ll \lambda$

1) $d \ll r \ll \lambda$ near zone (2)式で $e^{ikR} \simeq 1$ $\quad A(x,t) = \frac{\mu_0}{4\pi}\int_V dx' \frac{\mathbb{J}(x',t)}{R}$

2) $d \ll r \sim \lambda$ 中間帯

3) $d \ll \lambda \ll r$ 波動体 (wave zone) $kR \gg 1$

$r' = |x'| \sim d$ Rを展開できる

$R = |x - x'| \simeq r - \hat{n}\cdot x' \quad (\hat{n} = x/r)$

(2) → $A(x,t) = \frac{\mu_0}{4\pi}\int dx' \frac{\mathbb{J}(x',\omega)}{r - \hat{n}\cdot x'} e^{ik(r - \hat{n}\cdot x')}$

位相では $\hat{n}\cdot x'$ を残す

$$\cong \frac{\mu_0}{4\pi}\cdot\frac{e^{ikr}}{r}\int dx'\, \mathbb{J}(x',\omega)\, e^{ik\hat{n}\cdot x'} \qquad (15)$$

$\frac{d}{\lambda} \ll 1$ → 位相 $|k\hat{n}\cdot x'| \ll 1$

$$e^{-ik\hat{n}\cdot x'} = \sum_{n=0}^{\infty} \frac{(-ik)^n}{n!}(\hat{n}\cdot x')^n \qquad (16)$$

## §4 双極子放射 Dipole radiation

(16) → (15) $\quad n=0$ の項

$$A(x,\omega) \cong \frac{\mu_0}{4\pi}\cdot\frac{e^{ikr}}{r}\int dx'\, \mathbb{J}(x',\omega) = \frac{\mu_0}{4\pi}\cdot\frac{e^{ikr}}{r}(-i\omega)\int dx'\, \rho(x',\omega')\, x' \qquad (1)$$

$\int dx'\, \rho(x',\omega')\, x' = \mathbb{P}(\omega)$

$$\int dx\, \nabla\cdot\mathbb{J}(x,\omega)\, x_i = -\int dx\, J_i$$

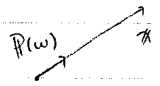
Eqn of continuity

$$-i\omega\rho(x,t) + \nabla\cdot\mathbb{J}(x,\omega) = 0$$

$$A = -i\frac{\mu_0\omega}{4\pi}P(\omega)\frac{e^{ikr}}{r} \qquad (2)$$

$$\mathbb{B}(\mathbf{x},\omega) = \frac{k^2}{4\pi\epsilon_0 c}\,\hat{\mathbf{n}}\times\mathbb{P}(\omega)\,e^{ikr}\,\frac{1}{r}\left(1-\frac{1}{ikr}\right) \qquad (3)$$

$$\mathbb{E}(\mathbf{x},\omega) = \frac{k^2}{4\pi\epsilon_0}\,e^{ikr}\,\frac{1}{r}\left\{(\hat{\mathbf{n}}\times\mathbb{P})\times\hat{\mathbf{n}} + \left[3\hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\mathbb{P})-\mathbb{P}\right]\frac{1}{kr}\left(\frac{1}{kr}-i\right)\right\} \qquad (4)$$




$kr \gg 1$ (wave zone)

$\mathbb{B}, \mathbb{E} \propto \frac{1}{r}$

$$\begin{cases} \mathbb{B} = \dfrac{k^2}{4\pi\epsilon_0 c}\,(\hat{\mathbf{n}}\times\mathbb{P})\,\dfrac{e^{ikr}}{r} \\ \mathbb{E} = c\,\mathbb{B}\times\hat{\mathbf{n}} \end{cases} \qquad (5)$$

Poynting vector $\mathbb{S}_F = \frac{1}{\mu_0}\,\mathbb{E}\times\mathbb{B}$

$$\mathbb{S}_F(\mathbf{x},t) = \frac{k^2 P^2(\omega)}{(4\pi\epsilon_0)^2\mu_0 c}\cdot\frac{\cos^2(kr-\omega t)}{r^2}\times\sin^2\theta\,\hat{\mathbf{n}} \qquad \cos\theta = \hat{\mathbf{n}}\cdot\mathbb{P}/P$$

## §5 磁気双極子

電気4重極放射

$n=1$

$$\mathbb{A}(\mathbf{x},\omega) = \frac{\mu_0}{4\pi}\,\frac{e^{ikr}}{r}\int d\mathbf{x}'\,\mathbb{J}(\mathbf{x}',\omega)\times(-ik)(\hat{\mathbf{n}}\cdot\mathbf{x}')$$

$$\mathbf{m} = \frac{1}{2}\int d\mathbf{x}'\,\mathbf{x}'\,\mathbb{J}(\mathbf{x}')$$

$$Q_{ij} = \int d\mathbf{x}'\,\rho(\mathbf{x}')\,(3x_i' x_j' - \delta_{ij} r'^2)$$

静電場 $\Phi_2(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{1}{2}\cdot\frac{x_i x_j}{r^5}\,Q_{ij}$

$$\boxed{\mathbb{A}_i(\mathbf{x},\omega) = \frac{\mu_0}{4\pi}\cdot\frac{e^{ikr}}{r^2}\left\{-ik\,(\mathbf{m}(\omega)\times\mathbf{x})_i - \frac{\omega k}{3r}\,Q_{ij}\,x_j\right\}}$$

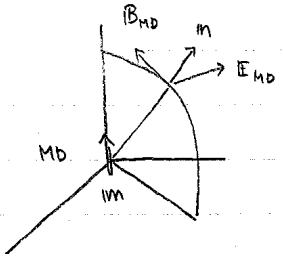
magnetic dipole

$m \longleftrightarrow \frac{1}{c}p$

$E_{MD} \longleftrightarrow cB_{ED}$

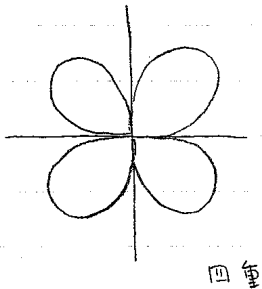
$cB_{MD} \longleftrightarrow E_{ED}$

角度分布は同じ




$B \propto \hat{n} \times \dot{p}$

強度 $\frac{\mu_0}{4\pi}\left(\frac{m^2\omega^4}{3c^3}\right)$



四重極子

中性子星 (Pulsar)

$M \simeq 0.5 \sim 1.5\, M_\odot$ $R \sim 10\text{km}$

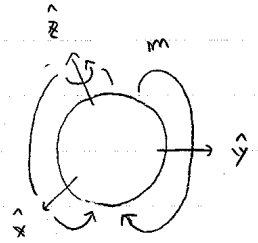
$2\pi/\omega = T = 0.088$ Vela

$0.033$ Crab

$\frac{dT}{dt} \sim 4.2 \times 10^{-13}$ (減衰)

$m_T = m_T(\cos\omega t\, \hat{x} + \sin\omega t\, \hat{y})$

$\frac{\mu_0}{4\pi}\frac{2m_T^2\omega^4}{3c^3}$

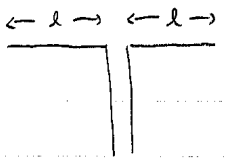



$B \sim 10^8\,T$

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## §6 Antenna

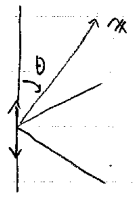



$$\vec{J} = \hat{z}\, I\, \delta(x)\delta(y)\sin k(l-|z|)\, e^{-i\omega t}$$

端で $\vec{J} = 0$

$$\vec{A}(\vec{x},\omega) = \frac{\mu_0}{4\pi}\frac{e^{ikr}}{r}\int d\vec{x}'\, \vec{J}(\vec{x}',\omega)\, e^{-ik\hat{n}\cdot\vec{x}'} \qquad \cdots (1)$$

$r \gg d$ $\quad \int d\vec{x}'\, \vec{J}(\vec{x}',\omega)\, e^{-ik\hat{n}\cdot\vec{x}'} = I\hat{z}\int_{-l}^{0} dz'\sin k(l-|z'|)\, e^{-ikz'\cos\theta}$

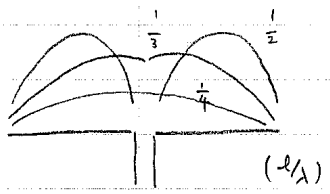



$$= \frac{2I\hat{z}}{k}\left[\frac{\cos(kl\cos\theta) - \cos kl}{\sin^2\theta}\right] \qquad \cdots (2)$$

エネルギー損失（時間平均） $W = \dfrac{\mu_0 c I^2}{8\pi^2}[\;''\;]\, d\Omega \qquad \cdots (3)$

放射抵抗（普遍定数） $R_{rad} = \dfrac{\mu_0 c}{2\pi}\displaystyle\int_0^{\pi} d\theta \sin\theta\,[\;''\;] = \frac{\mu_0 c}{2\pi} = 60\,(\Omega)$

Total Power $\quad \dfrac{1}{2} R_{rad} I^2 \qquad \cdots (4)$

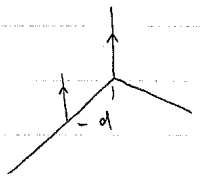



$l < \lambda$ $(l/\lambda \ll 1)$ になる ほど dipole の傾向

$l > \lambda$ multipole

phase の違..

2つの dipole $\quad \vec{J}(\vec{x}') = -i\omega\sum_i P_i\,\delta(\vec{x}'-\vec{d}_i)\, e^{i\beta_i}$




$$\int d\vec{x}'\, \vec{J}(\vec{x}')\, e^{-ik\hat{n}\cdot\vec{x}'} = -i\omega\sum_j P_j\, e^{-ik\hat{n}\cdot\vec{d}_j}\, e^{i\beta_j} = -i\omega P\hat{z}\,(e^{-ikd\sin\theta\cos\varphi}$$

$(\vec{d}_2 = \hat{x}d)$

$$\frac{dW}{d\Omega} = \frac{\mu_0}{(4\pi)^2}\frac{2\omega^4}{c}\sin^2\theta \times \cos^2\frac{1}{2}(kd\sin\theta\cos\varphi)$$

$kd = \pi$ のとき $\quad \theta = \dfrac{\pi}{2} \quad \varphi = 0 \quad \dfrac{dW}{d\Omega} = 0$

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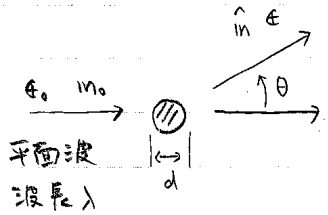
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## §7 長波長の電磁場の散乱

$$\mathbb{E} = \frac{k^2}{4\pi\varepsilon_0 r}(\hat{n}\times\mathbb{P})\times\hat{n}\, e^{ikr}$$

$$\mathbb{B} = \frac{1}{c}\hat{n}\times\mathbb{E}$$

} ・・・(1)




$\mathbb{P}$ : $\lambda$ 射波による分極

$\lambda$ 射波 直線偏光 $\epsilon_0$

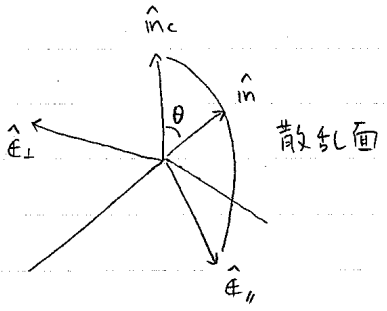
$\omega = kc$ $kd \ll 1$

$$d\sigma(\hat{n},\hat{\epsilon},\hat{n}_0,\hat{\epsilon}_0)/d\Omega = r^2|\mathbb{E}_{sc}|^2/|\mathbb{E}_{int}|^2 \quad \cdots(2) \qquad \dim\sigma = L^2$$

Dielectric sphere 半径 $a$. $\varepsilon$

$$\mathbb{P} = 4\pi\varepsilon_0 a^3 \frac{\varepsilon/\varepsilon_0 - 1}{\varepsilon/\varepsilon_0 + 2}\mathbb{E}_{inc} \quad \cdots(3)$$

$$\frac{d\sigma}{d\Omega} = k^4 a^6 \left(\frac{\varepsilon/\varepsilon_0 - 1}{\varepsilon/\varepsilon_0 + 2}\right)(\hat{\epsilon} - \hat{\epsilon}_0)^2 \quad \cdots(4)$$




$$\frac{\frac{d\sigma_\perp}{d\Omega} - \frac{d\sigma_{//}}{d\Omega}}{\frac{d\sigma_\perp}{d\Omega} + \frac{d\sigma_{//}}{d\Omega}} = \frac{\lambda^2\theta}{1+\cos^2\theta}$$

空の青色 分子 $\mathbb{P} = \alpha\mathbb{E}_{inc}$

$$\alpha = \frac{2\varepsilon_0}{N}(n-1)$$

$n$ 屈折率

$N$ 粒子数密度 ($m^{-3}$)

$$\bar{\sigma} = \int \sigma\, \lambda^2\theta\, d\theta = \frac{2}{3\pi}\cdot\frac{k^4(n-1)^2}{N^2}$$

attenuation constant $\eta$ Flux $\propto e^{-\eta x}$

$\eta = N\bar{\sigma}$ $\eta^{-1} \sim 10^2$ km

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# 第11章 運動する点電荷の作る電磁場

## §1 Lienard - Wiechert の potential

$\boldsymbol{x} = \boldsymbol{x}_0(t)$ given ... (1)

$$\left.\begin{aligned} \phi(\boldsymbol{x},t) &= \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\boldsymbol{x}', t-\frac{1}{c}|\boldsymbol{x}-\boldsymbol{x}'|)}{|\boldsymbol{x}-\boldsymbol{x}'|}\,d\boldsymbol{x}' \\ \boldsymbol{A}(\boldsymbol{x},t) &= \frac{\mu_0}{4\pi}\int \frac{\boldsymbol{j}(\boldsymbol{x}', t-\frac{1}{c}|\boldsymbol{x}-\boldsymbol{x}'|)}{|\boldsymbol{x}-\boldsymbol{x}'|}\,d\boldsymbol{x}' \end{aligned}\right\} \quad \cdots (2)$$

点電荷 $\rho(\boldsymbol{x},t) = q\,\delta(\boldsymbol{x} - \boldsymbol{x}_0(t))$ ... (3)

Retarded time $t_R = t - \frac{1}{c}|\boldsymbol{x} - \boldsymbol{x}_0(t_R)|$ ... (4) $t_R$で送った signal → $t$で $\boldsymbol{x}$ に至る

$$\phi = \frac{1}{4\pi\epsilon_0}\iint_{-\infty}^{\infty} \frac{\rho(\boldsymbol{x}',t')}{|\boldsymbol{x}-\boldsymbol{x}'|}\,\delta(t'-t+\tfrac{1}{c}|\boldsymbol{x}-\boldsymbol{x}'|)\,dt'\,d\boldsymbol{x}'$$

$$= \frac{q}{4\pi\epsilon_0}\iint_{-\infty}^{\infty} \frac{\delta(\boldsymbol{x}'-\boldsymbol{x}_0(t'))}{|\boldsymbol{x}-\boldsymbol{x}'|}\,\delta(\,\text{〃}\,)\,dt'\,d\boldsymbol{x}'$$

$$= \frac{q}{4\pi\epsilon_0}\int_{-\infty}^{\infty} \frac{1}{|\boldsymbol{R}(t')|}\,\delta(t'-t+\tfrac{1}{c}|\boldsymbol{R}(t')|)\,dt' \quad \cdots (5)$$

$\boldsymbol{R}(t) = \boldsymbol{x} - \boldsymbol{x}_0(t)$

$\boldsymbol{v}(t) = d\boldsymbol{x}_0(t)/dt$

$\boldsymbol{j} = q\,\boldsymbol{v}(t)\,\delta(\boldsymbol{x} - \boldsymbol{x}_0(t))$ より

$$\boldsymbol{A} = \frac{\mu_0}{4\pi} q \int_{-\infty}^{\infty} \frac{1}{|\boldsymbol{R}(t')|}\,\boldsymbol{v}(t')\,\delta(t'-t+\tfrac{1}{c}|\boldsymbol{R}(t')|)\,dt' \quad \cdots (6)$$

$$\int \delta(f(t))\,F(t)\,dt = F\left(\frac{df}{dt}\right)^{-1}\Big|_{f(t)=0}$$

$$\frac{d}{dt'}\left(t'-t+\frac{1}{c}|\boldsymbol{R}(t')|\right) = 1 - \frac{1}{c}\cdot\frac{\boldsymbol{R}(t')\cdot\boldsymbol{v}(t')}{|\boldsymbol{R}(t')|}$$

$$\left.\begin{aligned} \phi &= \frac{q}{4\pi\epsilon_0}\cdot\frac{1}{|\boldsymbol{R}(t_R)| - \frac{1}{c}\boldsymbol{R}(t_R)\boldsymbol{v}(t_R)} \\ \boldsymbol{A} &= \frac{\mu q}{4\pi}\cdot\frac{\boldsymbol{v}(t_R)}{|\boldsymbol{R}(t_R)| - \frac{1}{c}\boldsymbol{R}(t_R)\boldsymbol{v}(t_R)} \end{aligned}\right\} \quad \cdots (7)$$

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Lorentz 条件をみたす

$$\mathbb{E} = -\frac{q}{4\pi\epsilon_0}\nabla_R\int\left(\frac{1}{R}\delta(t'-t+\frac{1}{c}R)\right)dt' - \frac{\mu_0 q}{4\pi}\frac{\partial}{\partial t}\int\frac{v(t')}{R}\delta(t'-t+\frac{1}{c}R)\,dt' \qquad R=|\mathbb{R}(t')|$$

$$= -\frac{q}{4\pi\epsilon_0}\int\hat{R}\left(\frac{-1}{R}\delta(t'-t+\frac{1}{c}R)+\frac{1}{R}\frac{\partial}{\partial R}\delta(t'-t+\frac{1}{c}R)\right)dt' - \frac{\mu_0 q}{4\pi}\int\frac{\dot{v}(t')}{R}\frac{\partial}{\partial t}\delta(t'-t+\frac{1}{c}R)\,dt'$$

$$= \frac{q\hat{R}}{4\pi\epsilon_0}\cdot\frac{1}{R^2}\cdot\frac{1}{1-R\cdot v/cR}$$

$$\mathbb{E}_a = \frac{q}{4\pi\epsilon_0}\cdot\frac{\hat{R}\times\left[(\hat{R}-\frac{v}{c})\times a\right]}{c^2R(1-\frac{R\cdot v}{c})^3}\Bigg|_{t_R} \qquad \mathbb{E}_v = \frac{q}{4\pi\epsilon_0}(1-\frac{v^2}{c^2})\frac{\hat{R}-\frac{v}{c}}{R^2(1-\frac{R\cdot v}{c})^3}\Bigg|_{t_R} \qquad \cdots (10)$$

$\mathbb{E} = \mathbb{E}_a + \mathbb{E}_v$ $\quad \mathbb{E}_v \propto \frac{1}{R^2}$ $\quad \mathbb{E}_a \propto \frac{1}{R}$

$\mathbb{B} = \frac{1}{c}\hat{R}\times\mathbb{E}$ $\quad$ Poynting vector $\mathbb{S}_f = \frac{E^2}{\mu_0 c}\mathbb{R}$

$v \ll c$ なら $R\sim r$ $\quad \mathbb{E}_v = \frac{q}{4\pi\epsilon_0}\left[\frac{\hat{n}}{r^2}+\frac{1}{c^2r}\left[\hat{n}\times(\hat{n}\times a)\right]\right]$

$(\hat{n}\cdot a)\hat{n} - a$

## §2 放射減衰 (Radiation damping)

Non-Relativistic $\quad v \ll c$ $\quad \mathbb{S}_f \cong \frac{1}{\mu_0 c}\left(\frac{q}{4\pi\epsilon_0}\right)^2\frac{1}{c^4}\cdot\frac{1}{r}\left|\hat{n}\times(\hat{n}\times a)\right|^2$

$$\frac{dW}{d\Omega} = \left(\frac{q}{4\pi}\right)^2\frac{1}{\varepsilon_0 c^3}|a|^2\sin^2\theta \qquad W = \frac{2}{3}\cdot\frac{q^2|a|^2}{4\pi\epsilon_0 c^3} \quad (\text{単位時間あたり})$$

$W = \mathbb{F}_r\cdot v$

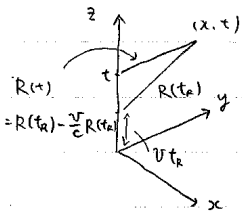
Relativstic $\quad W = \frac{2}{3}\cdot\frac{q^2}{4\pi\epsilon_0 c^3}\left[a^2-(v\times a)^2\frac{1}{c^2}\right]$

Synchrotron radiation $\quad a\sim\rho\omega^2\sim\rho\left(\frac{\rho}{v}\right)^{-2}$ $\quad W\sim\frac{q^2}{4\pi\epsilon_0 c^3}\cdot\frac{v^4}{\rho^2}$

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## §3 一様な運動




$v \parallel \hat{z}$

$x_0(t) = vt\hat{z}$

$t_R = t - \frac{1}{c}|x - x_0(t_R)|$ (1)

$R(t) = x - x_0(t) = x - vt\hat{z}$

$R(t_R) = x - vt_R\hat{z}$ (2) ( $R(x, t_R)$ の $x$ は省略 )

(1),(2) $R^2(t_R) = x^2 - 2vzt_R + v^2t_R^2$

$c^2(t-t_R)^2 = x^2+y^2+(z-vt_R)^2$ (3)

$ct_R = \gamma^2\left\{(ct-\beta z) \pm \sqrt{(ct-\beta z)^2 + \gamma^{-2}(r^2 - c^2t^2)}\right\}$

$\beta = v/c \qquad \gamma = 1/\sqrt{1-\beta^2}$

$-$ sign ( $\because \beta \to 0 \quad t_R = t - \frac{r}{c}$ )

§1 (7) $\phi(x,t) = \frac{q\gamma}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{x^2+y^2+\gamma^2(z-vt)^2}}$ ← $z' = \gamma(z-vt)$

N.R. $x' = x \quad y' = y \quad z' = z - vt$

§1 (10) $E_v = \frac{q}{4\pi\epsilon_0}\gamma^{-2}\frac{\hat{R} - v/c}{R^2(1-\frac{\hat{R}\cdot v}{c})^3}\Big|_{t=t_R}$

$B = \frac{1}{c}\hat{R} \times E$

### 相対性

B. E と場 net charge 0

電線 ⊕ イオン rest
⊖ 電子
← v

I = Nev N: N個の電子 / 単位長さ

q: test charge

系O : B だけ
系O' ( 速度 v で動く ) q は v で動く qv × B wire に向う力

} Para dox

### Lorentz 収縮

| | 長さ | |
|---|---|---|
| 棒の静止系 | $l_0$ | N本 / 単位長さ |
| v で動く系 O' | $l' = l_0\sqrt{1-\beta^2}$ | N本 / $\sqrt{1-\beta^2}$ |

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電子に対しては O′ が静止系

O′　N₋

O　$N_- / \sqrt{1-\beta^2}$ = N に等しい　　$N_- = \sqrt{1-\beta^2}\, N$

O′からみると net charge

$$\left( \frac{N}{\sqrt{1-\beta^2}} - N\sqrt{1-\beta^2} \right) e = \frac{Ne}{\sqrt{1-\beta^2}} \beta^2$$

O′のみる電流　$I = Ne v / \sqrt{1-\beta^2}$

∴ q の受ける力

① $qE' = q \frac{Ne}{\sqrt{1-\beta^2}} \beta^2 \frac{1}{2\pi\varepsilon_0 r}$

② $qv \times B = qv\, \mu_0 Ne v \frac{1}{\sqrt{1-\beta^2}} \cdot \frac{1}{2\pi r}$

$$\Bigg\} \quad \frac{qNe}{\sqrt{1-\beta^2}} \cdot \frac{v^2}{2\pi\varepsilon_0 r} \left( \frac{1}{c^2} - \varepsilon_0 \mu_0 \right)$$

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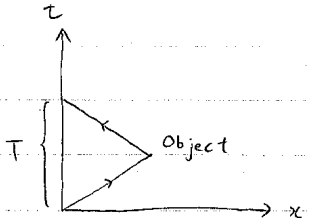
# 第12章 特殊相対論

1 距離、時間 not absolute but relative

原子時計

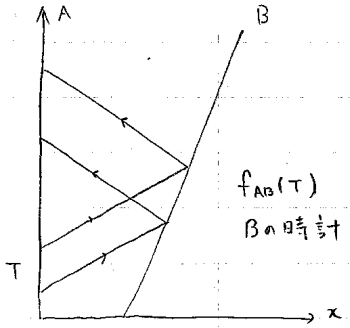
光

距離 $\frac{1}{2}cT$




$c=1$ とおく

k-factor

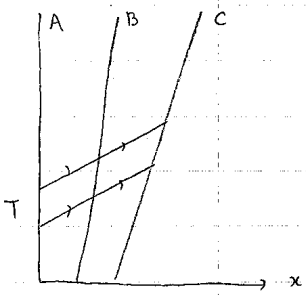



$f_{AB}(T)$ Bの時計

逆に $f_{BA}(T)$

$f_{AB}(T) = f_{BA}(T)$ Reciprocity

合成則

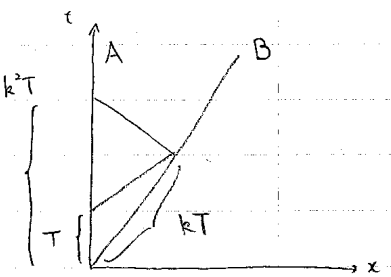



$f_{AC}(T) = f_{BC}(f_{AB}(T))$

もっと simple な関係 $f(T) = kT$

$k_{AC} = k_{AB}k_{BC}$

相対速度




Aが $t_1$ で signal を出し

〃 $t_2$ を受けとる

Aは Bが $\frac{1}{2}(t_1+t_2)$ に signal を受けとったと考える

$\frac{1}{2}(T+k^2T)$

Bがその間に動いた距離 $\frac{1}{2}(k^2T-T)$

$$v = \frac{k^2T-T}{T+k^2T} = \frac{k^2-1}{k^2+1} \quad \rightarrow \quad k = \sqrt{\frac{1+v}{1-v}}$$

$v \rightarrow -v$ ; $k \rightarrow \frac{1}{k}$

$$v_{AC} = \frac{k_{BC}^2 k_{AB}^2 - 1}{k_{BC}^2 k_{AB}^2 + 1} = \frac{v_{AB}+v_{BC}}{1+v_{AB}v_{BC}}$$

$|x|, |y| \leq 1$

$f(x,y) = \frac{x+y}{1+xy}$

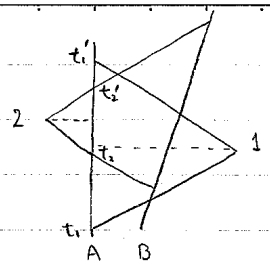
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## 同時性

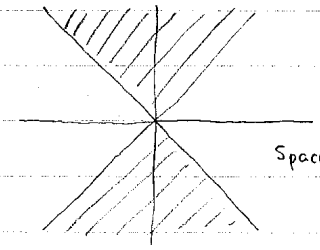
事象 event




(A) event 1 $\frac{t_1+t_1'}{2}$

event 2 $\frac{t_2+t_2'}{2}$

Aにとって同時 $t_1+t_1' = t_2+t_2'$

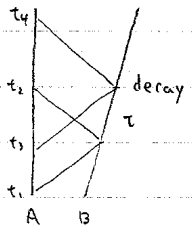


Space like

time like

Causality

## 時間のびる (Dilatation)




$\pi^+ \to \mu^+ + \nu$

life time $\tau$

co-moving B

$\frac{t_1+t_2}{2} \to \frac{t_3+t_4}{2}$

$\tau_{\pi^+} \cong 2.6\times10^{-8}$ sec

$\gamma \sim 2000$ $v \sim c$

$c\tau \sim 8m \to 16km$

$\tau_A = \frac{1}{2}(t_3+t_4-t_1-t_2)$

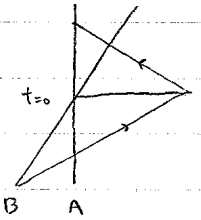
$t_4 - t_2 = k_{BA}\tau$

$k_{AB}(t_3-t_2) = \tau$ より

$\tau_A = \frac{1}{2}\left(k_{BA}\tau + \frac{1}{k_{AB}}\tau\right)$

$\therefore \tau_A = \frac{1}{\sqrt{1-v^2}}\tau \quad (=\gamma\tau)$

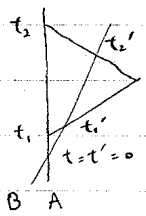
## Lorentz 収縮




## Lorentz 変換

ガリレイ $K \to K'$

$$\begin{cases} t = t' \\ x = x' + vt \end{cases}$$




$P(x,t)$ … A

$(x',t')$ … B

(A) $\begin{cases} t = \frac{t_1+t_2}{2} \\ x = \frac{t_2-t_1}{2} \end{cases}$

(B) $\begin{cases} t' = \frac{t_1'+t_2'}{2} \\ x' = \frac{t_2'-t_1'}{2} \end{cases}$

$t_1' = k_{AB}t_1$

$t_2' = k_{BA}t_2$

$\therefore$ $\boxed{\begin{aligned} t' &= \gamma(t-\beta x) \\ x' &= \gamma(x-\beta t) \end{aligned}}$

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A $dt$ $dx$

B $dt'$ $dx'$ $\quad dt'^2 - dx'^2 = \gamma^2\{(dt-\beta dx)^2 - (dx-\beta dt)^2\} = dt^2 - dx^2 \equiv ds^2$

Invariant

## 世界間隔

2つの event $\quad (x_1, y_1, z_1, t_1)$

$(x_2, y_2, z_2, t_2)$

$$l_{12}^2 = (x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2$$

$$S_{12}^2 = c^2(t_2-t_1)^2 - l_{12}^2 \quad (1)$$

time like $S_{12}^2 > 0$

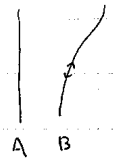
Space like $S_{12}^2 < 0$

$x_i = (x, y, z, ict)$

$$S_{12}^2 = -(x_{2i}-x_{1i})(x_{2i}-x_{1i})$$

$$ds^2 = -dx_i\,dx_i$$

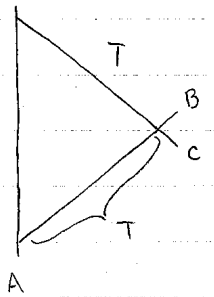
## 固有時間 Proper time




$$ds^2 = c^2dt^2 - (dx^2+dy^2+dz^2) = c^2dt'^2$$

$$dt' = \sqrt{1-\beta^2}\,dt$$

## 時計の paradox




宇宙船 $2T$

地球 $\left(k_{AB} + \frac{1}{k_{BA}}\right)T > 2T$

time と path dependent

## 3次元回転

$x_1x_2 \quad x_2x_3 \quad x_3x_1$

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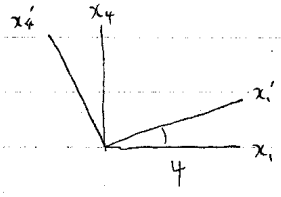
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$x_4 = ict$

$dx_1^2 + dx_2^2 + dx_3^2 + dx_4^2$

4次元 回転

$x_1x_4$ $x_2x_4$ $x_3x_4$




$x_1 = x_1' \cos\psi - x_4' \sin\psi$

$x_4 = x_1' \sin\psi + x_4' \cos\psi$ ψは相対速度 v

$P = (0, x_4)$

$\frac{x_1}{x_4} = \frac{v}{ic} = -\tan\psi$ $\begin{cases} \cos\psi = \eta \\ \sin\psi = i\eta\beta \end{cases}$

$\begin{cases} x_1 = \eta x_1' - i\eta\beta x_4' \\ x_4 = i\eta\beta x_1' + \eta x_4' \end{cases}$

$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$

$x = Sx'$

$S^T S = 1$ $S = \begin{pmatrix} \eta & 0 & 0 & -i\beta\eta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ i\beta\eta & 0 & 0 & \eta \end{pmatrix}$ 反対称テンソル

共変方程式 $A_{ij} = B_{ij}$

共変的な Maxwell eq

$J_i = (J_x\ J_y\ J_z\ ic\rho)$

$\partial_i J_i = 0$

$A_i = (A_x\ A_y\ A_z\ \frac{i\phi}{c})$

Lorentz 条件 $\partial_i A_i = 0$

$F_{ij} = \partial_i A_j - \partial_j A_i$ $\begin{pmatrix} 0 & B_z & -B_y & -\frac{1}{c}E_x \\ -B_z & 0 & B_x & \cdots E_y \\ B_y & B_x & 0 & \cdots E_z \\ \frac{1}{c}E_x & \cdots E_y & \cdots E_z & 0 \end{pmatrix}$

$\partial_j F_{ij} = \mu_0 J_i$

$\frac{1}{2}\epsilon_{ijkl}\partial_j F_{kl} = 0$